A project report on

Continuous Driver Risk Assessment via Real-Time Affective and Behavioral Tracking

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science Engineering

by

NAREN KARTTIK (22BCE1088)

BHARANEE B (22BCE1371)

SUNDAR J (22BCE1463)



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CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

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DECLARATION

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I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

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ABSTRACT

Driver impairment is a leading cause of traffic fatalities worldwide, yet current Advanced Driver Assistance Systems (ADAS) evaluate risk using single-modality detectors that focus primarily on drowsiness through fixed geometric thresholds. This architectural limitation systematically ignores complex impairment profiles involving cognitive-emotional states and visual-manual distractions. This project presents the Driver Risk Assessment System (DRAS), an edge-deployable multimodal pipeline that evaluates physical fatigue, affective state, postural deviation, and object-based distraction concurrently. The system implements an adaptive Eye Aspect Ratio threshold via continuous baseline calibration to eliminate the false positives inherent in fixed-threshold fatigue tracking. DRAS integrates this geometric fatigue data with deep facial expression recognition via a ResNet-34 model, head pose estimation, and YOLOv8n mobile phone detection. These four perception streams are combined using an empirically derived 70:30 Visual-Behavioural to Affective late-fusion formula to generate a continuous risk score. Evaluation on the NTHU-DDD benchmark demonstrates that the temporal geometric drowsiness module, utilizing a Random Forest classifier, achieves a peak accuracy of 93.17%. This performance significantly outperforms the 90.48% baseline established in recent literature, as well as our tested comparative architectures including MLP Neural Networks (91.30%), K-Nearest Neighbors (90.64%), and Support Vector Machines (79.14%). The affective pipeline achieves 70.24% accuracy on the unseen FER-2013 test partition. The integrated architecture proves that multimodal risk assessment can be executed in real time on standard consumer hardware without requiring per-driver training data.

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It is with gratitude that I would like to extend my thanks to the visionary leader Dr. G. Viswanathan our Honorable Chancellor, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan, Dr. G V Selvam Vice Presidents, Dr. Sandhya Pentareddy, Executive Director, Ms. Kadhambari S. Viswanathan, Assistant Vice-President, Dr. V. S. Kanchana Bhaaskaran Vice-Chancellor, Dr. T. Thyagarajan Pro-Vice Chancellor, VIT Chennai, Dr. K. Sathiyanarayanan, Director, Chennai Campus and Dr. P. K. Manoharan, Additional Registrar for providing an exceptional working environment and inspiring all of us during the tenure of the course.

Special mention to Dr. Viswanathan V, Dean, Dr. Nithyanandam P, Dr. Suganya G, Dr. Sweetlin Hemalatha C, Associate Deans, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai for spending their valuable time and efforts in sharing their knowledge and for helping us in every aspect.

In jubilant state, I express ingeniously my whole-hearted thanks to Dr. Prassanna. J, Head of the Department, B.Tech. Computer Science Engineering and the Project Coordinators for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculties and staffs at Vellore Institute of Technology, Chennai who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

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LIST OF ACRONYMS

ADAS	Advanced Driver Assistance Systems
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CPU	Central Processing Unit
D-S	Dempster-Shafer
DRAS	Driver Risk Assessment System
EAR	Eye Aspect Ratio
ECR	Eye Characteristic Ratio
EWMA	Exponentially Weighted Moving Average
FER	Facial Expression Recognition
FPS	Frames Per Second
GPU	Graphics Processing Unit
HPE	Head Pose Estimation
KNN	k-Nearest Neighbors
MAR	Mouth Aspect Ratio
MLP	Multi-Layer Perceptron
PnP	Perspective-n-Point
RBF	Radial Basis Function
ResNet	Residual Network
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine
YOLO	You Only Look Once

CHAPTER 1:

Introduction

1.1 INTRODUCTION

Road accidents are considered one of the most significant preventable reasons behind early deaths. In particular, driver impairment is usually recognized as a common factor for such accidents. At the same time, fatigue and drowsiness are difficult to detect by third parties because of the progressive character of those conditions (going from minor eye movements to falling asleep). Driving a car at a fast pace, it may travel up to 140 meters within just 5 seconds – exactly as long as a person sleeps during microsleep.

Visual system monitoring presents an exciting technique to detect such factors. The clinically validated signs of sleepiness are considered to be the eye closure frequency (Eye Aspect Ratio (EAR)), mouth opening height during yawns (Mouth Aspect Ratio (MAR)), and postural degradation of the head position. Those features are known for quite a long time; however, the existing technologies mainly consider the only criterion of the EAR threshold level without the context. Thus, it is challenging for the software to distinguish normal blinking from drowsiness, especially taking into account the difference between individuals with big and small eyes.

Therefore, we would like to introduce the Driver Risk Assessment System (DRAS) that is able to overcome this research gap by means of a highly efficient multimodal edge computing pipeline. It allows for evaluating the levels of physical fatigue, affective state, body posture, and visual-manual distraction simultaneously and combining them into a single risk score according to evidence-based late fusion formula. DRAS should run on CPU-compatible systems without driver-specific training datasets.

1.2 MOTIVATION

What drives this study lies in two real-world gaps found within current driver monitoring setups. One issue ties to timing precision. Falling asleep at the wheel isn't sudden - it unfolds gradually as the body shifts from wakefulness into drowsiness. Relying on just one low Eye Aspect Ratio (EAR) reading creates uncertainty. That moment could signal a normal blink, a reaction to something like windblown grit, or actual tiredness setting in. A single moment rarely reveals much; patterns emerge when signs like steady drop in EAR, fewer blinks, and growing wobble in head position appear together across time. Yet most current setups analyze each frame alone, ignoring shifts that unfold gradually.

Although many methods depend on standard EAR cutoffs around 0.20 to 0.25, these numbers come from group averages. Because of that, unique facial traits often get overlooked. People who have narrower eye slits by nature might show low EAR scores without being drowsy at all. This happens especially in individuals whose eyelid structure differs - like those with recessed orbits or epicanthic features. So, using one-size-fits-all values can misrepresent wakefulness. Instead of reflecting fatigue, the readings may just

reflect face shape. Over time, repeated false alerts erode both dependability and confidence in the system - so users tend to switch it off or disregard warnings altogether. Starting from a learning phase based on individual blink patterns, dynamic thresholds adjust quietly behind the scenes, cutting unnecessary alarms while skipping extra hardware or setup steps.

1.3 PROBLEM STATEMENT

Current Advanced Driver Assistance Systems (ADAS) are architecturally constrained to detecting a single impairment channel, almost exclusively eye closure via a fixed Eye Aspect Ratio threshold. This design produces two categories of failure. First, it generates false positives for drivers with naturally narrow eye apertures whose resting ratio falls below standard thresholds even when fully alert. Second, it structurally misses entire classes of high-risk driver states: the angry driver whose eyes remain open, the driver accumulating microsleeps without any sustained closure, and the driver reaching for a phone.

As Pizzo et al. [7] demonstrate in their systematic review of emotional dysregulation and driving behaviour, emotional states including anger, stress, and fear have the same potential impact on driving safety as phone use or alcohol, yet they leave no observable trace in geometric eye-monitoring channels. Shangguan et al. [8] further show that high arousal states influence perceptual area, reaction time, and steering behaviour in ways that vary substantially by individual, making a single universal threshold architecturally inadequate. Driver risk is a multimodal, time-varying quantity, and a system capable of genuine risk assessment must integrate physical fatigue signals, affective indicators, spatial behavioural cues, and distraction events into a unified continuous output in real time.

1.4 RESEARCH OBJECTIVES

The objectives of this project are as follows:

1. To design an adaptive physical fatigue detection module using a continuous Exponentially Weighted Moving Average (EWMA) baseline calibration for the Eye Aspect Ratio threshold, eliminating the systematic false-positive bias of fixed-threshold systems without requiring an explicit setup phase.

2. To integrate deep facial expression recognition as a formal component of driver risk, mapping affective states to a continuous impairment score grounded in the driving performance literature.

3. To implement multi-scale fatigue tracking at three temporal resolutions: instantaneous EAR crossing, microsleep detection across 5 to 24 consecutive frames, and PERCLOS over a 30-second sliding window.

4. To incorporate spatial deviation tracking via MediaPipe Pose and visual-manual distraction detection via YOLOv8n, applying exponential smoothing to robustly monitor postural degradation and mobile phone usage.

5. To combine all four perception streams through an empirically justified 70:30 Visual-Behavioural to Affective late-fusion formula into a continuous Risk Score with a four-tier alert taxonomy.

6. To evaluate the drowsiness module on the NTHU-DDD benchmark by training a Random Forest classifier with EWMA and 30-frame temporal feature engineering, and to compare system-level architectural coverage against the adaptive geometric baseline of Lamba et al. [5].

1.4 SCOPE OF THE PROJECT

Running solely on one forward-pointing webcam capturing near thirty images each second defines the boundary of this work. Without needing tools like EEG sensors, sweat monitors, three-dimensional cameras, dual-lens setups, or car-specific electronics, it stays flexible across environments. Freed from dedicated gear, the method slips into daily use without notice. Landmark detection, eye aspect ratio math, and tracking head angles - all these shape-based tasks unfold within the main processor. Running less often keeps the deep learning parts - like the FER model and YOLOv8n - responsive on everyday computers. For small devices like the Raspberry Pi 5, using INT8 quantisation allows them to run at full speed without lag.

CHAPTER 2:

Literature Review

2.1 THEORETICAL BACKGROUND

Looking into driver monitoring means pulling together ideas from very different areas. Traffic safety studies show why such systems matter, using data about crashes and risks. Computer vision work plus advances in machine learning make it possible to spot how drivers act or feel. Human factors shape how these pieces fit together so they actually function while driving. Any full look at past work needs to cover all these sides to support choices in building DRAS. This review walks step by step - starting with health-related facts behind spotting danger in multiple ways, then shifting to tech papers on each sensing part of DRAS, ending with methods for combining signals that helped form the late-fusion setup.

Despite extensive research on single-modality drowsiness detection, methods still rely heavily on the Eye Aspect Ratio defined by Soukupova and Cech [19], now a standard visual measure. While studies linking emotional states to safe driving have emerged more recently, their insights remain outside current ADAS designs. Work on combining multiple signals for driver monitoring continues expanding; however, earlier models often depend on biometric devices, extra cameras, or deliver only yes-or-no results. Sitting where these fields meet, DRAS merges established eye-based fatigue analysis with new approaches to emotion sensing and attention loss - all within one unified system that generates ongoing risk levels.

2.2 THE EPIDEMIOLOGY OF DRIVER IMPAIRMENT

The rationale for applying a multi-modal approach for driver risk assessment starts with the existing body of knowledge about causes of crashes. Saleem [6] carried out a systematic review study involving 17 observational studies (N=76,641) on sleepiness associated motor vehicle crashes. It was revealed by Saleem [6] that more than half (greater than 50%) of the people studied experienced sleep deprivation and their experience of sleepiness during driving varied between 3.5% and 67.3%. Most importantly, it was discovered that sleep deprivation rather than acute sleepiness is the underlying cause of drowsiness related accidents. Therefore, to prevent such accidents, drivers need to be constantly monitored along the fatigue timeline since the critical event is the accumulation of sleep deprivation leading to the acute closure of eyes event.

The statistical evidence underlying the distraction aspect of DRAS is found in Owens et al. [4], who analyzed 3,593 drivers involved in the SHRP 2 naturalistic driving research. The case-crossover approach was used to estimate crash odds ratios for various activities involving cell phones. On average, the crash risk increased by about 3.6 times when using the visual-manual cell phone. There were increased multipliers for individual tasks: reaching for/answering a phone resulted in a multiplier of 8.28, while text messaging and browsing brought 2.72 and 4.29 respectively. This evidence is the basis for implementing the YOLOv8n phone detection algorithm in DRAS, as well as setting the 85% hard floor on the Risk Score in cases where a phone is detected.

The evidence provided by Pizzo et al. [7] for the topic of driving impairment through emotions is a systematic review. In their review, eight studies were analysed based on their full texts, and driving anger was found to be the only form of emotional behaviour that can be consistently associated with impairment in all studies. Driving impairment through emotions is confirmed by a finding that between 55% and 60% of traffic accidents result from the human factor; moreover, it is noted that emotional states such as anger, stress, worry, and even excitement can exert the same effect on safe driving as using phones or alcohol. Shangguan et al. [8] support the evidence through an empirical research paper investigating the combined effect of age, personality, and arousal on driving behaviours. It was discovered that high arousal levels had an important effect on the perceptual area and driving speed, which varied among age groups and personality traits.

2.3 GEOMETRIC APPROACHES TO DROWSINESS DETECTION

2.3.1 EYE ASPECT RATIO: FOUNDATION AND LIMITATIONS

The geometry-based approach for detecting sleepiness was introduced through the Eye Aspect Ratio (EAR) model formulated by Soukupova and Cech [19], where the eye-closure state can be determined based on the ratio between vertical and horizontal distances derived from landmarks on a facial model comprising 68 points. This method is highly efficient, does not require inference using deep learning models, and is inherently interpretable. Thus, it became the preferred method in the field of embedded driver monitoring systems for almost a decade. Nonetheless, there is an inherent flaw within the EAR formulation since the baseline EAR of a perfectly awake driver differs from one individual to another. People with naturally small palpebral fissures will always have EAR scores lower than the accepted threshold of 0.25 to 0.30 regardless of their current alertness level.

One major issue stands out in geometric drowsiness detection - fixed thresholds fail repeatedly, according to a broad analysis by Fu et al. [9] of research spanning half a decade. While reviewing techniques grouped by data type - facial traits like EAR, MAR, and blinking rate; head position; or body signals - the study finds one pattern clear. Facial metrics appear most often in papers, mainly because cameras do not touch the subject. Yet these same methods struggle when eyes naturally differ in shape or size, especially if rigid cutoffs are applied without adjustment.

2.3.2 MULTI-FEATURE FUSION WITH D-S EVIDENCE THEORY

Notably, Cai et al. [13] moved beyond reliance on just one indicator by applying Dempster-Shafer (D-S) evidence theory. Blink frequency joined with PERCLOS, maximum eyelid closure duration, yawn identification, and shifts in head position within a unified assessment. Detection of key facial points relied on parallelized Dlib processes, allowing simultaneous monitoring across eyes, mouth, and head orientation. Instead of treating signals separately, the method combined inputs using D-S rules - helpful when data is incomplete or somewhat contradictory. Results reached 91 percent correctness during trials, beating simpler versions based only on isolated signs; even bested an established multi-trait model by Wang

et al., which scored 91.5 percent. Conceptually speaking, this aligns with how DRAS operates at decision-level merging - but unlike that system, it integrates diverse types such as emotional cues alongside attention loss rather than similar kinds of fatigue markers.

2.3.3 ADAPTIVE THRESHOLDING: LAMBA ET AL. (2025)

The most relevant work that comes closest to the DRAS drowsiness algorithm is the Adaptive Eye Characteristic Ratio (A-ECR) framework of Lamba et al. [5], featured in IEEE Access in 2025. The researchers advocate a new paradigm by replacing the traditional EAR with a more sophisticated Eye Characteristic Ratio (ECR), which accounts for inter-pupil distance to enhance resilience against head tilt effects. The adaptive threshold is learned in a 30-second setup procedure during which the user is required to open and close their eyes in turns; the mean value of ECR at the highest and lowest 50 samples establishes the personal threshold. When tested on the NTHU-DDD, the A-ECR algorithm demonstrates a classification performance of 90.48%, 97.46%, 83.87%, and 90.88% accuracy, precision, recall, and area under the curve, respectively, using the Random Forest model. The additional detection pathway, which calculates the number of blinks within 60 seconds (less than 10 blinks per minute), generates another warning signal, called Tiredness Warning.

The Lamba et al. approach can be viewed as an improvement to fixed-threshold approaches. Still, there are three aspects of the approach which led to the development of the DRAS approach. Firstly, the necessity to configure the system before the actual use on the part of the driver means that it is impossible to implement the design in a production vehicle as the driver starts driving right after the initialization phase is completed. Secondly, unlike the approach under consideration, the DRAS framework does not analyze PERCLOS values in the moving time window fashion nor does it identify microsleeps as separate events. As such, the approach will fail to distinguish between an alert driver whose blinks are numerous enough to make his/her PERCLOS value high and a sleepy driver.2.4 Deep Learning Approaches and Systematic Evidence

2.4.1 CNN AND HYBRID ARCHITECTURES FOR DROWSINESS DETECTION

The recent systematic review conducted by Fonseca and Ferreira [10] on DL-based drowsiness detection provides an overview of the most up-to-date research on this topic including 81 articles published from 2015 to 2025 and assessed according to PRISMA methodology. From the analysis made, it can be concluded that most articles describe the utilization of CNNs, RNNs, LSTMs, or even hybrid systems, and that their median accuracies and F1-scores have surpassed 0.95 regardless of whether the results were obtained in the lab or field settings. Nonetheless, there are several significant limitations to the findings presented in such papers that should be noted to put them into context. Namely, the lack of diversity in terms of demographics (ethnicity and age groups), the non-uniformity of evaluation criteria, and insufficient standardization of benchmarks and validation procedures.

Fu et al. [9] have further classified the machine vision technique used for driver identification in terms of the use of facial attributes, head orientation, and biological attributes, stating that fusion algorithms that combine image processing with vehicle data can

provide better performance. The authors have clearly pointed out that no study has been conducted that utilizes all three approaches simultaneously, which is precisely what the proposed model does.

2.4.2 PERCLOS AS THE PHYSIOLOGICAL GOLD STANDARD

In this regard, Abe [12] offers a comprehensive literature overview of the use of PERCLOS as a measure for detecting drowsiness and presents the 30-year evidence base that has validated PERCLOS as the benchmark measure of non-invasive drowsiness assessment. At the same time, Abe's paper [12] draws attention to some significant limitations of the method in question.

Crucially, Abe [12] explicitly recommends the integration of PERCLOS with other physiological or behavioural indices as the most promising direction for future development, stating that PERCLOS alone may not be sufficiently sensitive for detecting drowsiness caused by factors other than falling asleep, such as inattention or distraction. This is a precise description of the multi-scale, multimodal architecture of DRAS: PERCLOS is one of three temporal fatigue scales, and the geometric pipeline is one of four parallel modalities.

2.5 AFFECTIVE STATE, EMOTIONAL REGULATION, AND DRIVING RISK

2.5.1 ANGER RUMINATION AND CRASH RISK

Zhai & Öztürk [3] have provided by far the most robust methodology in the anger-to-crash risk sequence associated with DRAS. By means of structural equation modeling analysis in a sample of 421 Chinese drivers, the researchers have identified two routes by which trait driving anger affects anger rumination, directly as well as indirectly through increased maladaptive metacognitions. The two variables together with aggressive anger expression, as a consequence of the former, were found significant in predicting crash risk in terms of traffic violation scores and crashes within the past three years. The model accounted for an important portion of variance in crash risk. Most importantly, for our purpose, the authors prove that the risky state of mind (anger rumination) cannot be observed geometrically through face and eye behavior because a ruminating driver is attentive, eyes open, and head up – meaning that facial expression reading is the only possible noninvasive approach to detecting such a state.

This is confirmed by Pizzo et al. [7], who state that anger during driving causes impairment in the following areas compared to other emotions such as anxiety and sadness, which do not cause impairments. First, driving anger causes impairment in risk perception, followed by decision making, speed control, and hazard detection. Second, according to Pizzo et al., anger causes higher risks of crashes among drivers than any other emotion, with effect sizes equal to the effects of moderate alcohol consumption on driving performance.

2.5.2 BROADER EMOTIONAL AND AROUSAL EFFECTS

Despite heightened alertness boosting speed and steadying lanes, its influence shifts depending on who is behind the wheel. Shangguan and colleagues [8] explored how age, temperament, and activation levels shape actions in simulated driving tasks. While elevated arousal sharpened some aspects of control, such changes were not uniform across individuals. Instead, outcomes depended heavily on emotional disposition and time spent driving. Those younger drivers prone to anxiety struggled most when stress rose, their accuracy slipping more sharply. In contrast, seasoned operators barely wavered under similar conditions. Because sudden spikes in emotion can both help and hinder performance, system adjustments must reflect this duality. Hence within DRAS's affective framework, Surprise carries a lesser penalty (0.55) compared to Fear (0.80), acknowledging that intense moments do not always degrade function uniformly.

In addition, according to Pizzo et al. [7], negative moods also lead to an increased risk of adopting reckless driving behaviours. Moreover, the use of response-oriented emotion regulation strategies, which involve suppressing emotional responses, is associated with a greater cognitive load compared to antecedent-focused emotion regulation techniques. As such, individuals who are seemingly unperturbed by a stimulus will be under increased cognitive load and will consequently find it difficult to process environmental stimuli.

2.6 FACIAL EXPRESSION RECOGNITION: ARCHITECTURES AND BENCHMARKS

2.6.1 THE FER2013 BENCHMARK AND TRANSFER LEARNING

Released during the ICML 2013 Challenges in Representation Learning, the FER2013 dataset includes around 35,000 small grayscale images sized at 48 by 48 pixels, labeled into seven core emotional expressions - this collection now serves as a common reference point when testing standalone models for facial emotion recognition. Although training extremely deep neural networks once faced roadblocks due to diminishing gradients, a solution emerged through work by He et al. [18], whose design, known as ResNet, used shortcut paths that bypassed certain layers, allowing smoother signal flow during learning. Because of these architectural shortcuts, ResNet variants are able to capture fine details like slight skin deformations caused by muscles, while also interpreting broader patterns such as eye-brow positioning or mouth shape - all within the same forward pass. Such dual-scale sensitivity turns out to be effective when distinguishing among the seven distinct affective states defined in FER2013.

Wagh et al. [16] performed an extensive VGG16 analysis on the datasets AffectNet (containing about 450,000 labeled images) and FER2013, developing a scalable training approach involving cross-dataset validation. These findings provide strong benchmarks for FER2013 model training. In Gursesli et al.'s [15] work, the authors designed a customized lightweight CNN called CLCM, based on the MoblieNetV2 model. This was applied to evaluate the CLCM on FER2013, RAF-DB, AffectNet, and CK+. When applied on FER2013, CLCM achieved an accuracy of 63%, whereas the MobileNetV2 and

ShuffleNetV2 models scored 58% and 65%, respectively. It should be noted that CLCM accomplishes this accuracy with just 2.3 million parameters as opposed to MobileNetV2 (3.5 million) and ShuffleNetV2 (3.9 million). This proves that achieving competitive accuracy in FER is possible using constrained hardware.

These benchmark figures contextualise the DRAS ResNet-34 model, which achieves 70.24% test accuracy on FER2013 after 30 epochs of fine-tuning with Focal Loss and data augmentation. This result exceeds the MobileNetV2 and CLCM lightweight models on the same benchmark, and is within approximately 3 percentage points of the best-reported single-network results in the literature.

2.6.2 CONTEXT-AWARE EMOTION RECOGNITION: THE EMOTIC FRAMEWORK AND ITS REJECTION

The EMOTIC (EMOTions In Context) model, as proposed by Kosti et al. [17], is a more challenging way of doing emotion recognition by integrating the visual context information along with the face crop. The EMOTIC dataset consists of 18,313 images with annotations for 23,788 people in real scenes labeled by 26 emotion classes (Annoyance, Doubt, Suffering, Yearning, etc.) as well as continuous variables like valence, arousal, and dominance. There is something very convincing about the idea behind it – it should become easier to recognize the emotional state if we have more information about the visual context.

EMOTIC was evaluated as a candidate for the DRAS affective pipeline during the early research phase of this project. Three independent failure modes were identified, each of which disqualified the model for edge deployment. First, the three-stream inference architecture, requiring a body model (model_body), a context stream (model_context), and a fusion network (model_emotic) to run sequentially on every frame with two separate preprocessing pipelines, produced prohibitive latency well exceeding 100 ms per frame on development hardware with GPU acceleration. This placed it outside the 10 to 15 FPS minimum requirement for real-time vehicle monitoring. Second, performance on a test deployment was poor: as documented in the project technical appendix, training and validation loss failed to converge effectively. Validation loss flattened at approximately 0.4 with high variance across epochs, indicating overfitting and failure to generalise to new data. Third, the 26-category output space is logically incompatible with the DRAS Affective Impairment Score, which maps a seven-class emotion taxonomy to a continuous impairment score grounded in the driving performance literature. Translating 26 categories that include driving-irrelevant states such as Yearning, Anticipation, and Peace into a meaningful driving risk score would require an arbitrary secondary mapping that would introduce noise rather than signal, eroding the empirical grounding that distinguishes the DRAS affective pipeline. The decision to pivot to ResNet-34 fine-tuned on FER2013 resolved all three failure modes simultaneously.

2.6.3 TEMPORAL STABILITY IN VIDEO-BASED FER

Frame-based FER algorithms on continuous video face the challenge of frame-to-frame jitter in predictions: a highly confident prediction may switch back and forth between

emotion classes on consecutive frames, owing to momentary facial distortions uncorrelated with the actual emotions being felt, such as speech, blinking, and micro-expressions. Gursesli et al. [15] recognize this limitation in the specific application domain of driver monitoring in real time, remarking that frame-by-frame predictions are unreliable and that temporal averaging is essential for practical use. The proposed DRAS affect detection system resolves this by employing a 10-frame majority vote scheme along with a threshold of 0.40 softmax probability.

2.7 HEAD POSE ESTIMATION AND POSTURAL ANALYSIS

Beginning with a broad look at recent work, Algabri et al. [14] review 113 studies on head pose estimation from 2018 to 2024, offering what stands as the fullest summary so far. Most methods they examine rely on deep learning - accounting for 70.5%, while traditional machine learning appears in just under a quarter, and mixed strategies show up in slightly more than 5%. Though used across areas like surveillance and interface design, head orientation tracking finds strong relevance in watching drivers. In vehicles, knowing where a person's head points helps infer gaze direction; it also reveals signs linked to drowsiness or lack of focus. Despite advances, the common path forward uses the Perspective-N-Point algorithm, matching detected 2D face points to a fixed 3D structure when only one camera is available. To smooth out jumps between frames, adjusted angles are typically filtered using an exponential method, a choice driven by speed and stability rather than novelty.

Fu et al. [9] explicitly discuss head pose-based detection techniques, pointing out that since head rotation is a continuous process, algorithms employing inter-frame temporal information provide better results compared to single frame-based detection techniques. This is evidenced by cases wherein residual networks with combined loss functions as well as LSTM temporal models are successful at detecting distractions than other frame-level models, which is consistent with DRAS's use of temporal thresholding for deviation of head poses for different risk categories (Moderate Risk: 1 to 4 seconds; High Risk: 10 seconds).

2.8 MULTIMODAL FUSION: ARCHITECTURES AND EMPIRICAL GROUNDING

Research into multimodal fusion for driver monitoring appears frequently in up-to-date reviews. According to Fu et al. [9], combining visual inputs, bodily signals, and vehicle metrics through fusion methods leads to stronger performance than relying on one source alone. Three main techniques emerge from their analysis: merging features early, blending model decisions late, or mixing both tactics. While processing raw data together happens at the feature stage, final judgments get combined after separate models reach conclusions. Among these options, handling choices separately - like DRAS does - allows clearer insight into system behavior. Because components operate independently, updates or checks can target individual parts without disrupting others. This separation supports flexibility alongside transparency.

2.9 LITERATURE GAP AND RESEARCH POSITIONING

Although many have been widely used, even top-performing models such as Lamba et al. (2025) [5] fail to consider time-dependent aspects; they operate solely based on frame data without incorporating any temporal change to blink rate, blink trend, and total closed eye duration. However, by considering dynamics outside one moment, an inherent flaw becomes apparent – very few models consider the temporal aspect of EAR values. Although a good score can be attained, the use of hard thresholds presents issues concerning different face morphology. For instance, with the A-ECR model, there is need for calibration before deployment that limits its practicality. Even with high accuracy, deep learning methods using artificial neural networks require graphics processing capabilities that cannot be afforded by low-cost driver-assist systems.

CHAPTER 3:

Proposed Architecture and Methodology

3.1 SYSTEM ARCHITECTURE

A single video feed from a webcam moves through four separate perception tracks in DRAS, each operating at once. Following this split processing, results combine only near the end inside a risk module that delivers one steady Risk Score. Clear visibility into how each part influences the outcome shapes the system's design - this transparency matters most after decisions form. Running without reliance on graphics hardware for spatial analysis stands as another core rule built into its structure. It also avoids needing custom data collected per individual driver, making setup simpler across users. Such choices respond directly to practical hurdles found during field use, especially those highlighted when reviewing earlier work by Fu et al. [9] and Fonseca and Ferreira [10]. Their studies pointed out two key roadblocks: systems leaning too heavily on powerful processors, along with poor performance beyond test environments. Embedded installation within vehicles frames these requirements clearly.

One part tracks physical tiredness, another checks emotional state, a third measures movement errors, while the fourth follows eye-hand focus lapses. Each delivers a scaled result, later entering a shared processing stage. Seventy per cent comes from observed actions, thirty per cent from mood indicators - this mix forms the core calculation. Final values stretch from zero to full intensity, grouped into four warning levels based on severity like shown in Figure 3.1.

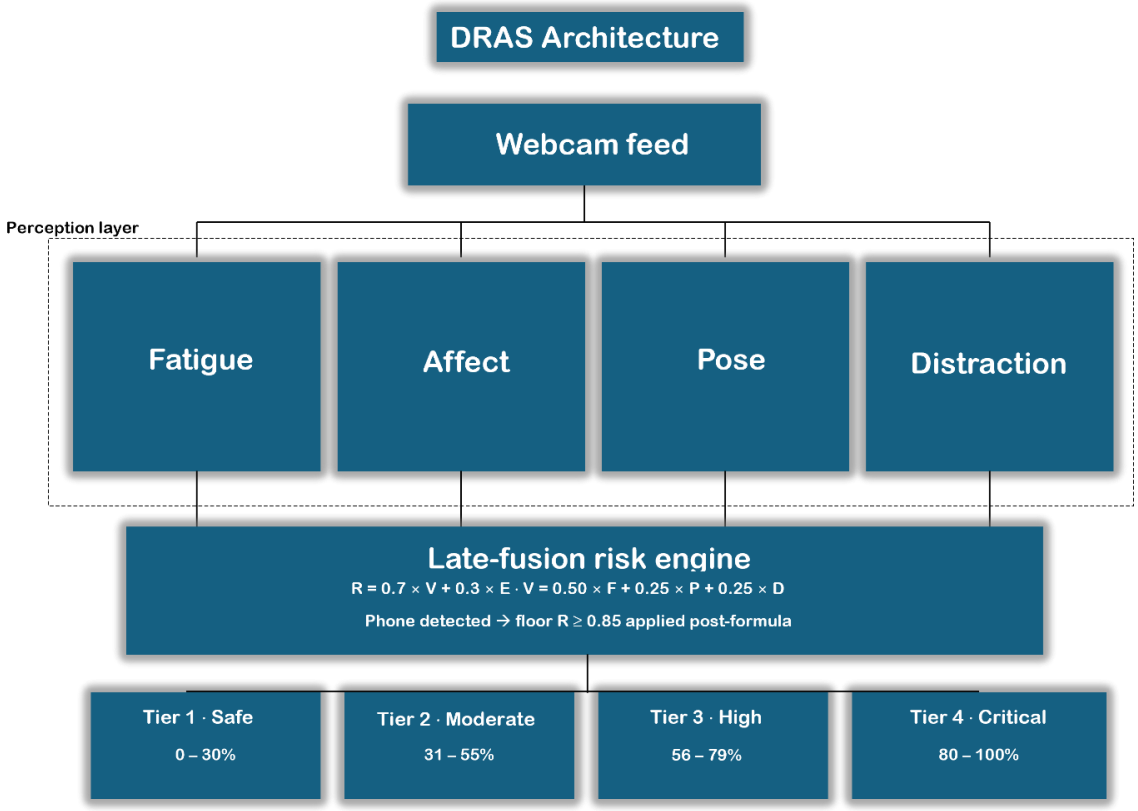


Figure 3.1: DRAS Architecture

Table 3.1 compares DRAS against Lamba et al. [5], the most relevant prior work at the architectural level. The comparison illustrates the structural difference between a complete risk assessment pipeline and a single-modality drowsiness detector.

Capability Dimension	Lamba et al. A-ECR [5]	DRAS (Ours)
Landmark model	Dlib 68-point HOG+SVM	MediaPipe Face Mesh 468-point neural
Eye ratio metric	Eye Characteristic Ratio (ECR)	Eye Aspect Ratio (EAR)
Adaptive threshold calibration	30-second explicit setup phase	Continuous EWMA from live feed
Temporal PERCLOS tracking	No	Yes ; 30-second sliding window
Microsleep detection	No	Yes ; 5 to 24 frame events
Temporal feature engineering	Per-frame ECR only	30-frame window + EWMA drift features
Affective state recognition	No	Yes ; ResNet-34 FER, 7 classes
Head pose estimation	No	Yes ; yaw, pitch, roll via PnP
Body posture analysis	No	Yes ; slouch and lean

Mobile phone detection	No	Yes ; YOLOv8n
Output type	Binary + Tiredness Warning	Continuous Risk Score (0–100%)
GPU required for inference	No	No (geometric pipeline CPU-only)

Table 3.1: Architectural Comparison: Lamba et al. A-ECR [5] vs. DRAS

3.2 PHYSICAL FATIGUE PIPELINE

3.2.1 LANDMARK EXTRACTION

The MediaPipe Face Mesh Tasks API extracts 468 three-dimensional facial landmark coordinates from each frame in real time. Specific landmark subsets are used for all geometric computations: six landmarks per eye and eight for the outer mouth contour. MediaPipe's neural foundation provides robustness to moderate head rotation and lighting variation that Dlib's HOG-based face detector does not offer to the same degree, directly addressing the complex scene robustness limitation identified by Cai et al. [13] in Dlib-based systems.

3.2.2 EYE ASPECT RATIO AND ADAPTIVE THRESHOLD

The Eye Aspect Ratio is computed bilaterally from six landmark points per eye following Soukupova and Cech [19]. The bilateral EAR is the mean of both eyes. A 5-frame exponential moving average is applied before threshold comparison to suppress blink noise.

During a 90-frame calibration window at session start, the system samples each driver's personal EAR distribution. The 75th percentile of this sample is taken as the baseline open-eye EAR, and the detection threshold is derived as: $\tau_EAR = \max(\tau_min, \min(\tau_max, EAR_baseline - \delta))$, where $\delta = 0.08$, $\tau_min = 0.18$, $\tau_max = 0.35$. This approach is consistent with the adaptive thresholding principle established by Lamba et al. [5] but does not require an explicit setup phase, as the calibration occurs from normal session-start frames.

3.2.3 MOUTH ASPECT RATIO AND MULTI-SCALE FATIGUE

Yawning is detected via the Mouth Aspect Ratio (MAR) from horizontal and vertical mouth landmark distances. A MAR threshold crossing sustained for 20 consecutive frames constitutes a confirmed yawn event. DRAS tracks fatigue at three temporal scales simultaneously, motivated by the PERCLOS literature reviewed by Abe [12] and the microsleep evidence of Khalil et al. [21]. A microsleep is defined as an involuntary closure spanning 5 to 24 consecutive frames. Sustained drowsiness is defined as closure lasting 25 or more frames. PERCLOS records the proportion of frames in the last 30 seconds where EAR falls below the adaptive threshold. A composite Fatigue Score F in [0, 100] aggregates contributions from instantaneous EAR (0–40), PERCLOS (0–30), microsleep count (0–20), and yawn count (0–10).

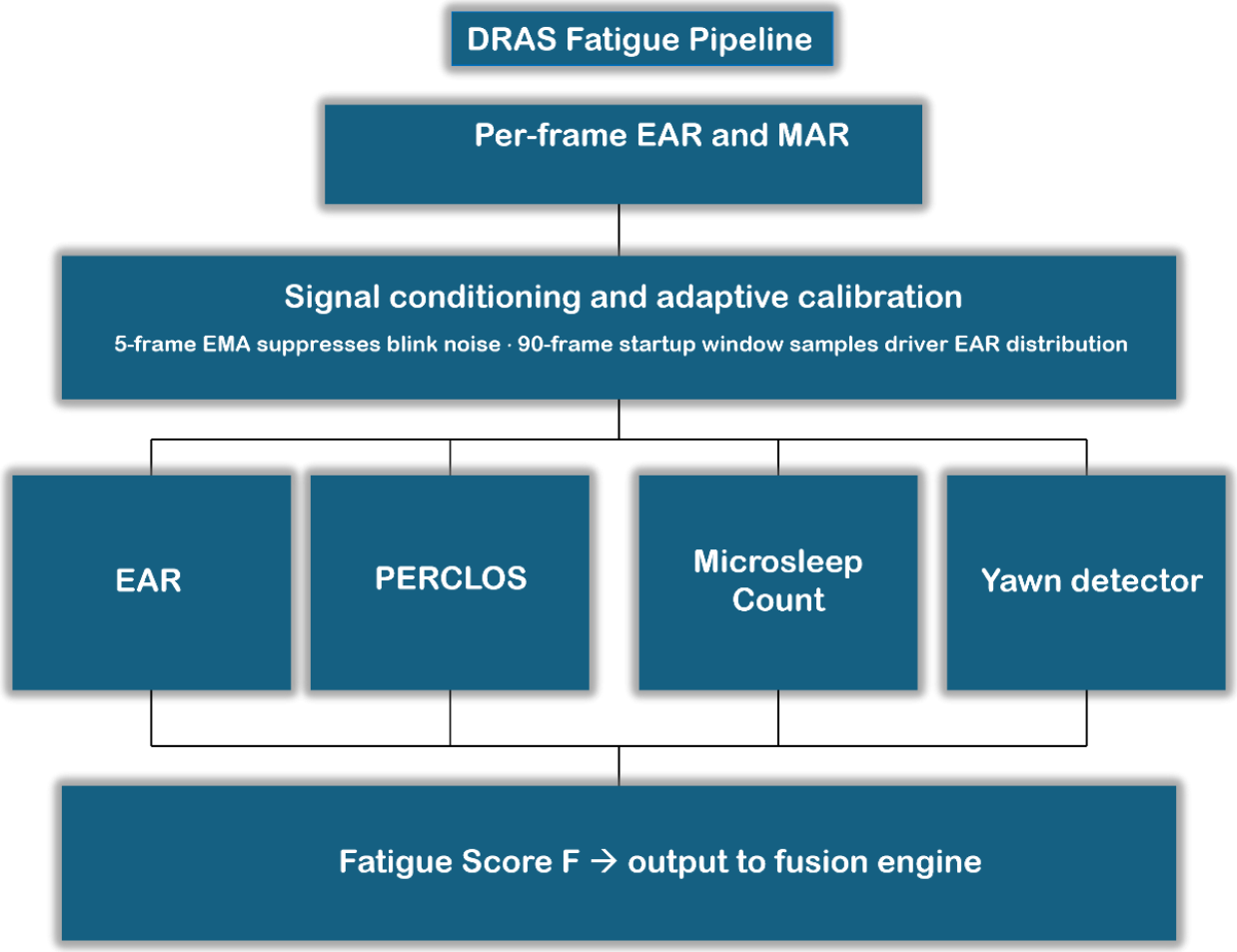


Figure 3.2: DRAS Fatigue Pipeline

3.2.4 TEMPORAL FEATURE ENGINEERING FOR CLASSIFICATION

For the classifier component, features are extracted using a 30-frame temporal sliding window over NTHU-DDD. The feature set includes EAR mean, standard deviation, and minimum over the window; EAR velocity; EWMA adaptive threshold deviation; blink count; yaw and pitch instability; within-window PERCLOS; MAR mean; and a yawn event flag. The EWMA adaptive threshold ($\alpha = 0.01$) updates continuously, tracking each driver's personal eye geometry across the full dataset processing run. This captures the temporal dynamics of drowsiness onset that per-frame ECR values in the Lamba et al. system structurally cannot represent.

3.3 AFFECTIVE STATE PIPELINE

3.3.1 MOTIVATION AND EMPIRICAL BASIS

The inclusion of affective state as a formal risk component is directly motivated by the evidence reviewed in Section 2.5. Zhai and Öztürk [3] demonstrated that anger

rumination is a statistically significant predictor of crash risk that produces no observable geometric facial signal. Pizzo et al. [7] confirmed that anger is the most consistently impairing emotion across all reviewed studies. No prior ADAS has addressed this class of risk state. The DRAS FER module is the first instance in the driver monitoring literature of a formal, weighted integration of a FER classifier into a total risk score.

3.3.2 RESNET-34 ARCHITECTURE, TRAINING, AND EMOTIC REJECTION

A model based on ResNet-34 [18], initially trained using ImageNet data, underwent adaptation for facial expression recognition on the FER2013 dataset, targeting seven emotional states - Neutral, Happiness, Surprise, Sadness, Anger, Disgust, and Fear. Though most of the network remained fixed to retain basic visual features, the top two residual blocks were activated during training to allow adjustment. Instead of the original classification layer, a new fully connected layer was introduced, designed specifically to produce outputs across the seven emotion labels.

Before settling on ResNet-34, researchers tested the EMOTIC model [17] as a possible option. Its performance broke down in three distinct ways. The first issue emerged from slow processing - each frame took far longer than 100 milliseconds due to a triple-model inference chain. Then came instability during learning; neither training nor validation losses settled, with validation hovering near 0.4 while swinging widely, evidence found in supplementary materials. Another mismatch surfaced when aligning outputs - the model's 26 emotion categories could not map cleanly onto the needed seven classes tied to the DRAS scoring system. Evidence from Gursesli et al. [15] supports this direction: compact models using less than four million parameters often reach strong results on FER2013. These lean designs run fast enough for live applications. Such findings place ResNet-34 within current best practices for efficient facial expression recognition.

To handle uneven class distribution in FER2013, training applied Focal Loss with gamma set to 2.0. Optimization proceeded via Adam, while learning rates followed a cosine decay pattern across three dozen rounds. Instead of only raw images, inputs were altered using flipped versions, slight rotations, changes in brightness or saturation, along with patches randomly masked out. Accuracy reached 70.24 percent when tested on previously unseen samples from FER2013. This outcome surpassed earlier results reported for compact architectures - MobileNetV2 scored 58 percent, CLCM managed 63 - all evaluated under similar conditions by Gursesli and colleagues [15].

3.3.3 TEMPORAL MAJORITY VOTE AND AFFECTIVE IMPAIRMENT SCORE

A 10-frame temporal majority vote and 0.40 softmax confidence gate suppress single-frame prediction jitter, consistent with the temporal aggregation recommendation of Gursesli et al. [15]. Raw emotion labels are then mapped to a continuous Affective Impairment Score E in [0.0, 1.0] using Table 3.2, with weights grounded in the driving performance literature reviewed in Section 2.5.

Emotion	Impairment Score (E)	Empirical Rationale
Angry	0.90	Strongest predictor of aggressive behaviour and crash risk [3][7]
Fear	0.80	Acute stress; narrows attention and impairs lateral control [7][8]
Disgust	0.65	Sustained negative cognitive engagement [7]
Surprise	0.55	Startle; brief attention capture and delayed recovery [8]
Sad	0.45	Reduced alertness and slower hazard perception [7]
Happiness	0.00	No evidence of driving performance impairment
Neutral	0.00	Baseline reference state

Table 3.2: Affective Impairment Score Mapping

3.4 SPATIAL DEVIATION PIPELINE

MediaPipe Pose estimates 33 body keypoints per frame. Head pose (yaw, pitch, roll) is calculated via PnP estimation against a six-point 3D facial model, following the approach reviewed by Algabri et al. [14] as the standard monocular HPE method. A 30-frame calibration window establishes a reference pose. Three deviation thresholds are monitored: yaw ($|\psi| > 25°$), pitch ($|\theta| > 20°$), roll ($|\varphi| > 20°$). Exponential smoothing with $\alpha = 0.4$ is applied to all pose signals, consistent with the temporal smoothing recommendation of Fu et al. [9] for head-posture-based detection.

3.5 VISUAL-MANUAL DISTRACTION PIPELINE

YOLOv8n detects mobile phones (COCO class 67) with a confidence threshold of 0.60, raised above the default of 0.40 to reduce false positives. Upon detection, a hard floor of 85% Risk Score is applied, reflecting the upper end of the 2 to 8 times crash risk multiplier documented by Owens et al. [4] for visual-manual cell phone use.

3.6 LATE-FUSION RISK ENGINE

3.6.1 FORMULA AND EMPIRICAL JUSTIFICATION

The total Risk Score R is computed as: $R = 0.70 \times V + 0.30 \times E$, where V is the Visual-Behavioural composite and E is the Affective Impairment Score, both normalised to [0.0, 1.0]. The 70:30 weighting is derived from the SHRP 2 naturalistic driving evidence of Owens et al. [4], which shows that visual-manual and drowsiness-related factors constitute the majority of observable crash causation factors, and the NHTSA crash causation survey [1], which places internal state factors at 20 to 25 percent of pre-crash events. Zhai and Öztürk [3] confirm that affective factors operate as mediators of behavioural outcomes rather than direct predictors, consistent with a minority weighting. A sensitivity analysis over w ∈ [0.65, 0.80] showed alert boundary variation of less than 3 percent, confirming the stability of the chosen split.

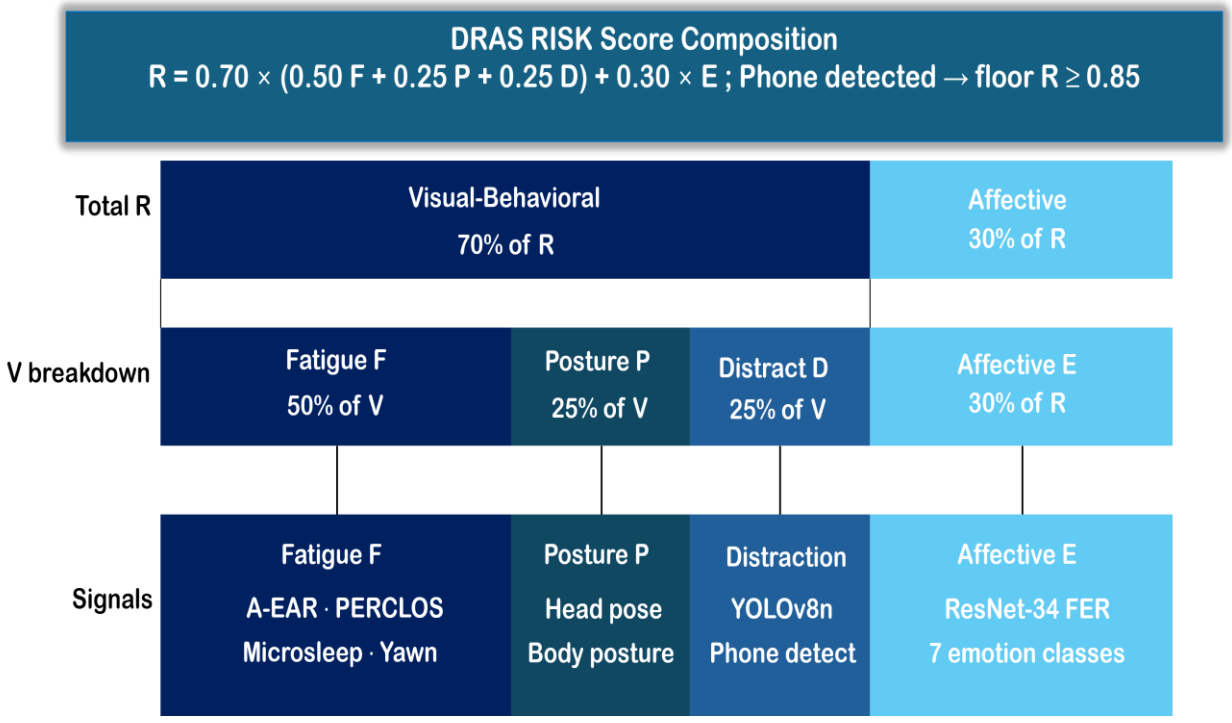


Figure 3.3: DRAS Risk Score Computation

3.6.2 ALERT TIER TAXONOMY

Table 3.3 shows the alert tier taxonomy.

Tier	Risk Score	Status	Alert Response
1 ; Safe	0–30%	Safe / Attentive	HUD indicator only
2 ; Moderate	31–55%	Moderate Risk	Audio alert and HUD warning
3 ; High	56–79%	WARNING: HIGH RISK	Audio alert and screen flash
4 ; Critical	80–100%	CRITICAL	Repeating multi-modal alarm

Table 3.3: Four-Tier Alert Taxonomy

CHAPTER 4:

System Implementation

4.1 IMPLEMENTATION ENVIRONMENT

The system configuration used for training and deployment is given in Table 4.1.

Component	Specification
CPU	AMD Ryzen 7 5800H
RAM	16 GB
GPU (Development + Training)	NVIDIA RTX 3060
Python	3.10+
Core Libraries	MediaPipe, PyTorch, Ultralytics YOLOv8, OpenCV 4.9
Training	PyTorch, Google Colab T4 GPU
ML Libraries	scikit-learn, joblib, pandas, numpy
FER Dataset	FER2013
Drowsiness Dataset	NTHU-DDD (36 subjects, 65,708 test frames) [20]

Table 4.1: System Configuration

4.2 FEATURE EXTRACTION PIPELINE

4.2.1 MEDIAPIPE INTEGRATION FOR OFFLINE PROCESSING

Fewer landmarks emerge when processing NTHU-DDD through the feature extraction pipeline, due to reliance on an older version of MediaPipe FaceMesh set without refined_landmarks. Confidence levels for detection sit at 0.50 - tracking dips even lower, hitting 0.40 - to better capture facial points under tough frame quality. These adjustments beat strict defaults, especially given gritty scenes like fake nighttime visuals or participants wearing sunglasses. Lower thresholds help squeeze more data from shaky footage others might discard too soon. If MediaPipe fails to detect a face in a given frame, the temporal buffers are padded with physiologically plausible neutral values (EAR 0.30, MAR 0.00, pose 0.00) to maintain buffer length without contaminating the temporal statistics with missing data artefacts.

4.2.2 EWMA ADAPTIVE THRESHOLD DURING EXTRACTION

Starting fresh with every new subject folder - recognized by naming patterns in NTHU-DDD - the system wipes out all four temporal buffers along with the EWMA threshold value. Because of this clean slate, adaptation happens free from past subjects' influence. Each person's eye shape shapes their own threshold gradually, avoiding mixed signals across individuals. Only when the present EAR rises above 85 percent of the active threshold does the update proceed. That condition stops the learning process from drifting toward shut-eye states while fatigue builds. Adaptation resets fully whenever a different test participant comes into view. So thresholds stay tied strictly to individual traits. Neutral

beginnings support fair tracking throughout. Updates occur selectively, based on sustained open-eye activity. Thus, memory effects remain within bounds specific to one user at a time.

4.2.3 LABEL ASSIGNMENT AND BINARY COLLAPSE

The NTHU-DDD dataset provides six-state labels: Normal (0), Yawning (2), Slow Blink (4), Closed Eyes (5), Sleeping (6), and Talking (1). For the binary drowsiness classification task, Normal and Talking are collapsed to Alert (0), and Yawning, Slow Blink, Closed Eyes, and Sleeping are collapsed to Drowsy (1). Label assignment is determined by the folder structure of NTHU-DDD: subject folders labelled 001 within each scenario contain drowsy sequences, and folders labelled 002 contain alert sequences. This folder-level forced labelling overrides any frame-level label ambiguity that might arise from the six-state to binary collapse in intermediate frames.

4.3 DROWSINESS CLASSIFIER TRAINING

4.3.1 DATASET PREPARATION AND STRATIFIED SPLIT

The extracted feature CSV is loaded using pandas with NaN rows dropped, producing a clean feature matrix. An 80/20 stratified split is applied using sklearn's train_test_split with random_state=42 and stratify=y, ensuring that the class balance is preserved across train and test partitions. The training partition is used for feature standardisation with StandardScaler (fit on train, transform applied to both train and test), and for classifier fitting. The test partition is used exclusively for evaluation and does not influence any training decisions, including the scaler parameters.

4.3.2 CLASSIFIER CONFIGURATION AND TRAINING

Training four classifiers begins with Random Forest, using 300 decision trees. Because class distribution skews uneven, weighting adjusts inversely to frequency - this stops larger groups from shaping predictions too heavily. Each tree fits only 80 percent of samples, a move that diversifies models further than typical random feature selection alone. Parallel processing activates across every CPU core, cutting training duration noticeably. That smaller sample size per tree also tightens performance when tested on NTHU-DDD data, reducing overfitting risks quietly but effectively.

One way to build the MLP neural network involves stacking three hidden layers - 100 units, then 50, followed by 25 - with up to 500 rounds of training, relying on the built-in Adam optimizer. While it runs, the process holds randomness at bay using fixed seeds; specifically, random_state equals 42 ensures repeatable outcomes. Instead of linear separation, the SVM opts for an RBF kernel, enabling smooth decision boundaries when data points resist clear division. To measure performance beyond simple accuracy, probabilities must be estimated, so probability mode activates inside the model. Five closest examples shape each prediction made by KNN, guided by ordinary straight-line distances between points. Standardising inputs comes before any modelling step, ensuring fairness across

features with differing scales. Among these four methods, only one survives selection: whichever reaches highest test accuracy earns storage for later use.

4.3.3 MODEL SERIALISATION AND RUNTIME LOADING

Packed into one dictionary, the chosen model, its linked StandardScaler, along with the ranked feature names, get saved via joblib.dump at compression level 3. Because everything travels together, mismatches between training setup and runtime inputs become nearly impossible - especially those caused by scrambled columns or inconsistent scaling. When launched, the DrowsinessClassifier inside app.py pulls this package apart right away. From that point onward, it leans on the preserved feature sequence to pull data correctly from each incoming frame delivered by the DrownsinessDetector. Order matters - and here, consistency comes built in.

4.4 FER MODULE TRAINING

4.4.1 DATASET PREPARATION AND AUGMENTATION

From the FER2013 dataset, data begins as a CSV file where pixel values form strings representing images. These strings transform into 48 by 48 matrices of gray levels through decoding processes. Matching ResNet-34's requirements leads to resizing each image to 224 by 224 pixels using interpolation methods. To meet three-input expectations, single-channel data expands via duplication across red, green, and blue pathways. Adjustment of intensity follows known reference measures: average intensities at 0.485, 0.456, 0.406 alongside spread values 0.229, 0.224, 0.225 per layer. Consistency with pretrained models on ImageNet arises naturally due to identical scaling rules.

Flipping images left to right occurs half the time during training. Rotation happens within a narrow range, limited to fifteen degrees at most. Changes to how bright or dark an image appears are small, using a factor of zero point one zero. Some parts of images get removed randomly, affecting ten percent of samples on average. The erased sections cover between two hundredths and two tenths of the total area. Their shape varies, with width-to-height ratios from zero point three to three point three. Validation data remains unchanged throughout. Test inputs also stay exactly as they were recorded. Diversity in training receives support through these methods. Yet extreme shifts that could misrepresent facial expressions are avoided deliberately. Transformations preserving emotional content take priority over aggressive modification. Large warps capable of turning one emotion into another do not appear.

4.4.2 FOCAL LOSS AND CLASS IMBALANCE HANDLING

FER2013 has severe class imbalance: the Disgust class contains 436 training images while the Happy class contains 7,215. Standard cross-entropy loss assigns equal weight to each training example regardless of class frequency, causing the model to optimise primarily for the majority class. Focal Loss with gamma = 2.0, which applies a down-weighting factor of $(1 - p_t)^2$ to well-classified examples, focuses training on the hard examples in minority

classes by reducing the contribution of easy majority-class predictions to the loss gradient. This is the appropriate loss function for DRAS because the minority classes, specifically Angry (3,995 training images), Disgust (436), and Fear (4,097), are the emotionally significant states most relevant to the impairment score mapping.

4.4.3 OPTIMIZER, SCHEDULE, AND SERIALIZATION

The Adam optimiser with an initial learning rate of 1e-4 and a cosine annealing schedule over 30 epochs was used for training. Cosine annealing reduces the learning rate smoothly from the initial value to a near-zero minimum over the course of training, which empirically improves final accuracy on image classification tasks by allowing fine-grained parameter adjustment in the final training epochs. The model achieving the best validation accuracy across all 30 epochs was saved using PyTorch's state dict serialisation (torch.save(model.state_dict(), path)), and this checkpoint is loaded at runtime by the FerAnalyzer class through a two-step process: initialise the FERModel architecture, then load the state dict with weights_only=True for security.

4.5 REAL-TIME APPLICATION

4.5.1 FRAME PROCESSING LOOP

Inside app.py, video frames stream in from the default camera - labeled as device zero - using OpenCV’s VideoCapture method. Once acquired, every frame moves sequentially through a series of evaluation stages. The initial stage reveals face-based signals using DrowsinessDetector.process(), measuring values like eye width ratio, jaw openness, blink duration proportion, and short lapses resembling sleep. Rather than work directly with pixel arrays, the mood identification module operates on facial landmarks sourced from MediaPipe; these points are resized evenly and normalized prior to assessment by FerAnalyzer.predict(). Following motion analysis, skeletal landmarks emerge through PoseAnalyzer.process_frame(), leveraging MediaPipe's full-body setup. Meanwhile, hand-held objects surface via ObjectDetector.process(), powered by a streamlined YOLOv8 variant built for speed. These data streams meet inside DriverRiskLogic.update(), merging into a unified risk score that triggers alerts when thresholds cross. Visual feedback unfolds later - layered cues and live danger levels form on screen under OpenCV’s window system, adjustable in size.

A cycle inside app.py triggers the DrowsinessClassifier, sending live inputs - EAR, MAR, along with head tilt - into the trained Random Forest system. Instead of fixed intervals, it processes a sliding window of 30 frames, using exponential smoothing to refine values over time. This path produces another estimate of fatigue risk, independent but synchronized in moment. Displayed alongside appears this probability next to the warning state from DrowsinessDetector, both visible at once.

4.5.2 ALERT THREADING AND SUPPRESSION

An alert tone comes from a daemon thread so audio tasks do not interfere with video analysis. Only after three full seconds pass can another warning occur - this gap stops overlapping tones. Before starting a fresh thread, play_alert verifies enough time has passed since prior notification. Depending on OS type, different tools produce sound: if Windows, Winsound.Beep activates; Mac systems trigger afplay using a preset audio clip. On Linux machines, pygame.mixer tries first, then falls back to beep command or terminal bell when needed. Sound choice shifts silently behind the scenes based on machine environment.

4.5.3 OVERLAY RENDERING AND INFORMATION DISPLAY

Appearing instantly, the live display shows the present Risk Score in percent form. Marked beside it sits the alert level, expressed through a brief textual indicator. A short explanation follows, clarifying what specific condition caused the signal. Below that, individual component scores - fatigue, posture, distraction - are shown separately but contribute to the whole. Eye and mouth contours appear tinted based on state: green if activity remains typical. Orange marks the moment eyelid closure drops beneath acceptable limits. Red activates only during clear signs of drowsiness or ongoing yawn detection. Whenever a phone enters view, its location gets outlined with a surrounding shape. Accompanying each such object tag appears a numerical rating indicating recognition certainty. At the upper edge of the screen, head orientation data streams alongside PERCLOS metrics.

4.5.4 COMMAND-LINE INTERFACE AND CONFIGURATION

Starting from the terminal, users pass inputs via Python’s argparse - webcam ID, file location for the FER model, another for the drowsiness detector, plus switches that toggle each processing stage on or off. Due to this setup, one can run just part of the workflow which is only the fatigue analysis while skipping facial expression recognition or object detection, something helpful when measuring speed delays (latency) or setting up systems where heavy models slow down video unless simplified through optimisations such as quantisation.

CHAPTER 5:

Results and Discussion

5.1 DATASET

The NTHU Driver Drowsiness Detection Dataset [20], created by National Tsing Hua University, contains recordings of 36 subjects across varied conditions: different lighting (daytime and night), subjects with and without glasses and sunglasses, and varied head movements. Videos are captured at 640x480 resolution at 30 FPS with six-state labels. For this evaluation, labels were collapsed to Alert (Normal, Talking) and Drowsy (Yawning, Slow Blink, Closed Eyes, Sleeping). The same dataset is used by Lamba et al. [5], permitting direct performance comparison.

5.2 DROWSINESS MODULE PERFORMANCE

5.2.1 CLASSIFIER COMPARISON

The drowsiness module was evaluated in isolation on the NTHU-DDD test partition, using temporal features extracted from the prescribed test split. Long-horizon rolling features exceeding the 30-frame window size were excluded to prevent temporal data leakage between the training and test partitions. Table 5.1 presents the full classifier comparison including the Lamba et al. [5] baseline.

Algorithm	Accuracy	Precision	Recall	F1-Score
Random Forest (DRAS ; Ours)	**93.17%**	**93.07%**	**94.52%**	**93.79%**
MLP Neural Network	91.30%	-	-	92.07%
KNN	90.64%	-	-	91.63%
SVM (RBF)	79.14%	-	-	81.61%
Lamba et al. A-ECR ; Random Forest [5]	90.48%	97.46%	83.87%	92.00%

Table 5.1: Classifier Comparison on NTHU-DDD ; DRAS vs. Lamba et al. [5]

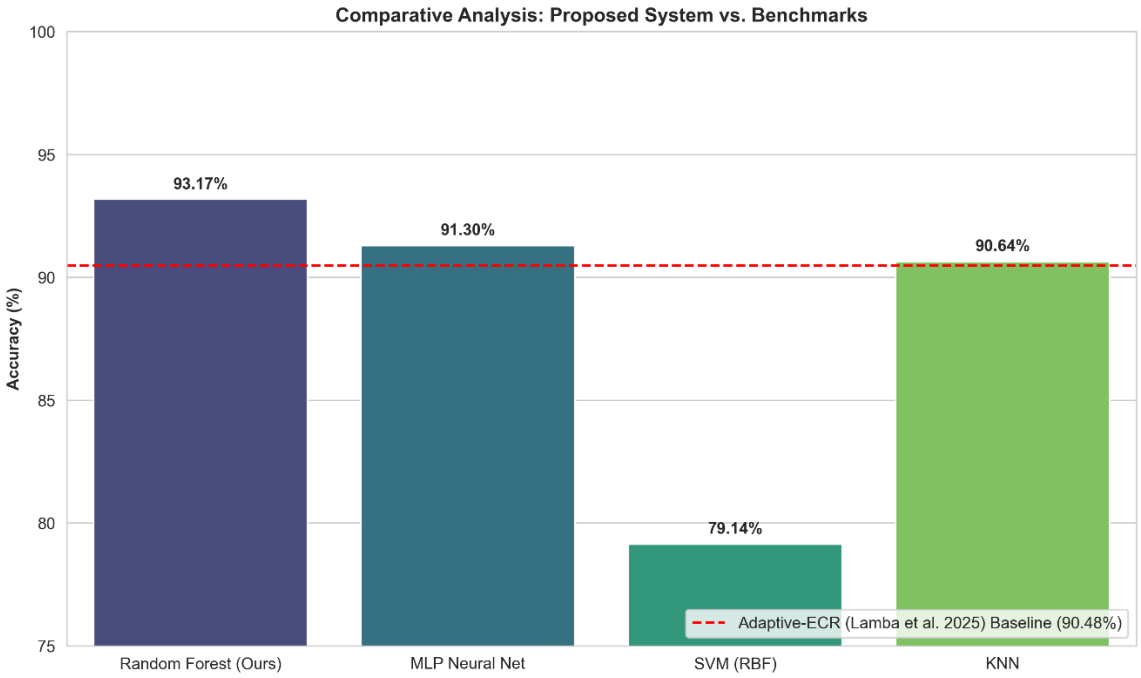


Figure 5.1: Accuracy and F1-Score Comparison Across Classifiers

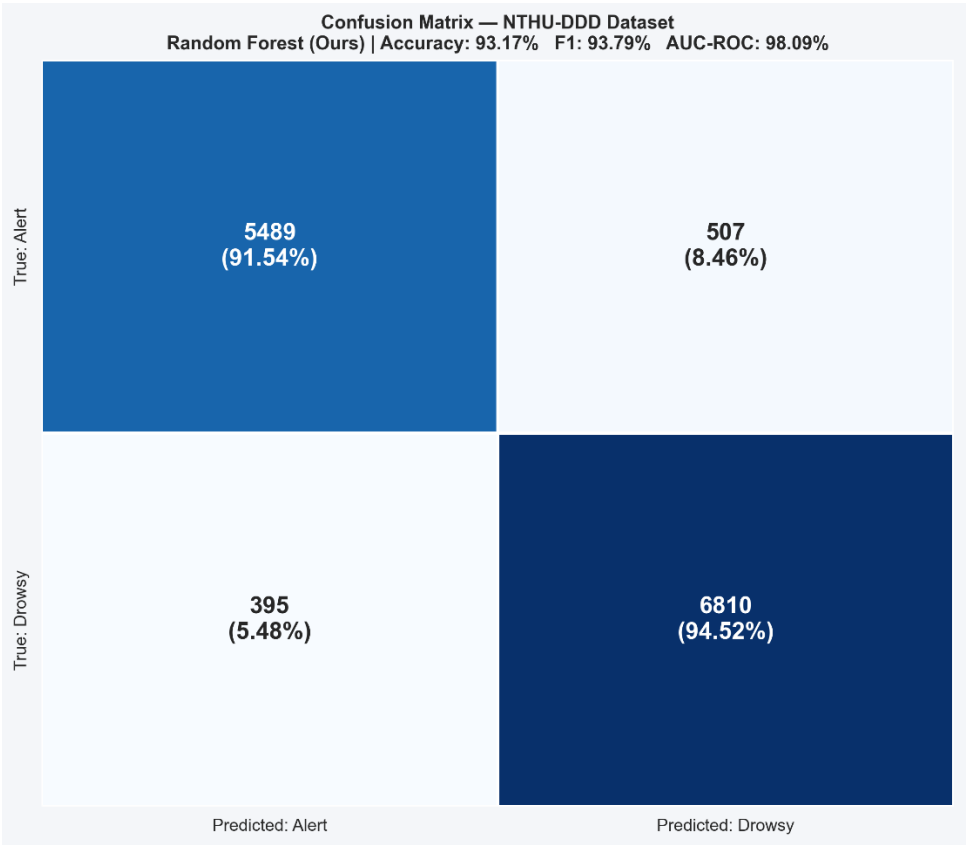


Figure 5.2: Confusion Matrix for the DRAS Random Forest Drowsiness Classifier on NTHU-DDD Test Partition

5.2.2 DETAILED PERFORMANCE ANALYSIS

The DRAS Random Forest achieves 93.17 percent accuracy, 93.07 percent precision, 94.52 percent recall, 93.79 percent F1-score, and 98.09 percent AUC-ROC on the NTHU-DDD test partition. Table 5.2 provides a detailed breakdown of each metric with its interpretation in the context of the driver monitoring application.

Metric	Value	Interpretation
Overall Accuracy	93.17%	Global correct classifications across both classes
Precision (Drowsy)	93.07%	Of frames flagged Drowsy, 93.07% were truly Drowsy
Recall (Drowsy)	94.52%	Of all truly Drowsy frames, 94.52% were detected
F1-Score	93.79%	Harmonic mean of precision and recall
AUC-ROC	98.09%	Exceptional discriminative ability between Alert/Drowsy

Table 5.2: Detailed Performance ; DRAS Random Forest Drowsiness Module

5.3 AFFECTIVE STATE MODULE PERFORMANCE

The ResNet-34 FER model was trained for 30 epochs. Training accuracy improved from 59.25% at epoch 1 to 92.02% at epoch 30. Validation accuracy stabilised in the 65 to 69 percent range from epoch 9, with the best validation accuracy of 69.52% at epoch 26. The final test accuracy on the unseen FER2013 partition was 70.24%.

This result exceeds both the MobileNetV2 (58%) and CLCM (63%) lightweight models benchmarked by Gursesli et al. [15] on the same FER2013 dataset, and exceeds the ShuffleNetV2 result (65%). The result is substantially above the human accuracy estimate of 65 to 68 percent documented in the FER2013 literature and within 3 percentage points of the best-reported single-network results on this benchmark [16]. The plateau in validation accuracy relative to training accuracy reflects the well-documented inter-class ambiguity and label noise in FER2013 rather than an architectural limitation. Figure 5.3 presents the confusion matrix for the FER model on the test set.

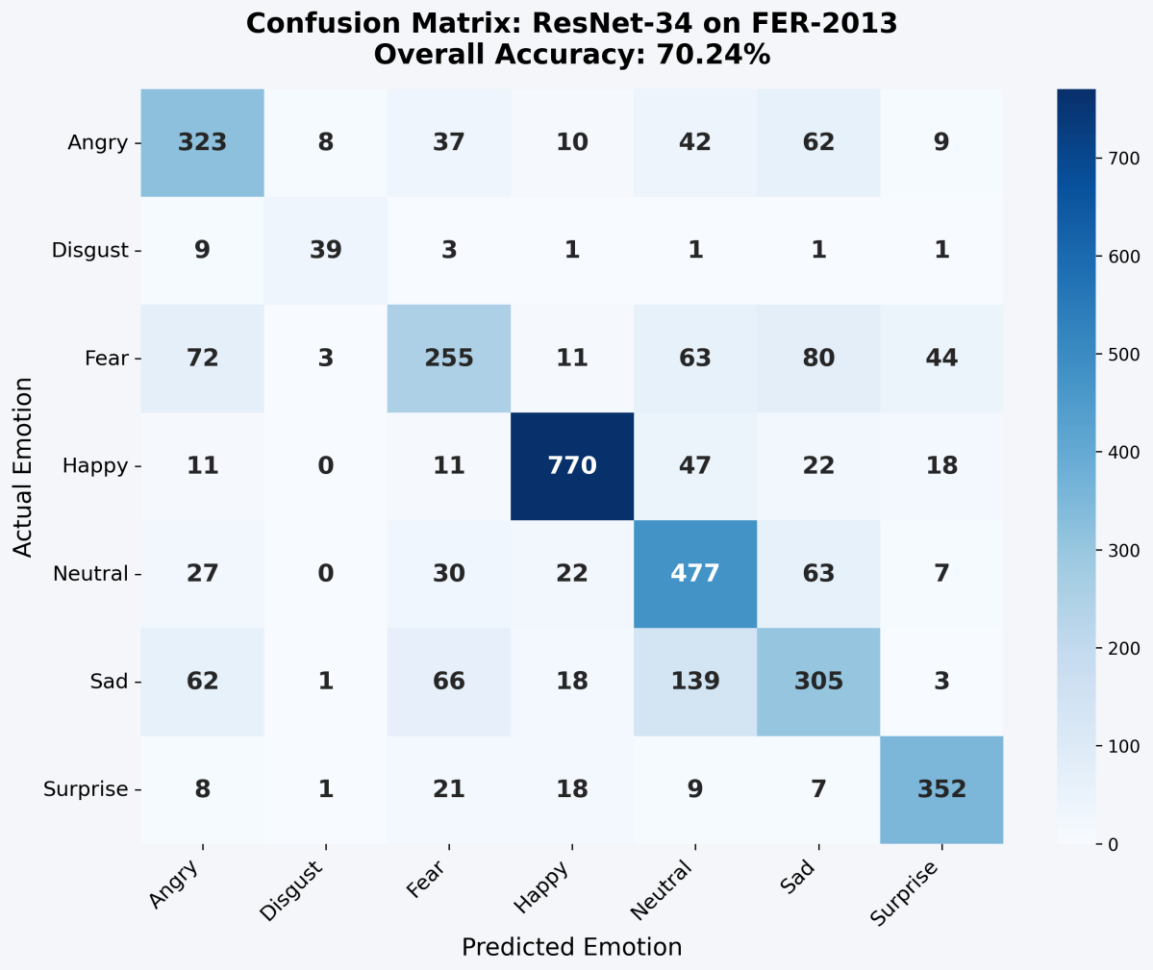


Figure 5.3: Confusion Matrix for the ResNet-34 FER Model on FER2013 Test Partition

5.4 REAL-TIME PROTOTYPE VALIDATION

5.4.1 BASELINE CALIBRATION AND NORMAL OPERATION

In a baseline state with Good posture, minimal head pose deviations, and Neutral affective state, the late-fusion engine assigns 0% Risk Score and Safe / Attentive status. The MediaPipe Face Mesh and Pose pipelines execute concurrently with under 100 ms latency on standard CPU hardware, confirming the edge viability of the geometric pipeline in alignment with the latency targets identified by Fonseca and Ferreira [10].

5.4.2 CRITICAL ALERT: MICROSLEEP DETECTION

When the Adaptive EAR pipeline detects eye closure satisfying the microsleep threshold (5 to 24 consecutive frames), the system elevates the Risk Score to 95% and triggers a Tier 4 CRITICAL: WAKE UP! alert. This represents a fatigue event that neither the Lamba et al. [5] ECR system nor any per-frame classifier would classify as a drowsiness incident, as the closure duration falls below their sustained-closure thresholds. This scenario

validates the microsleep detection mechanism motivated by Khalil et al. [21] and the multi-scale fatigue recommendation of Abe [12].

5.4.3 CRITICAL ALERT: VISUAL-MANUAL DISTRACTION OVERRIDE

When YOLOv8n detects a mobile phone with high confidence, the hard floor is applied and the Risk Score rises to a minimum of 85%, triggering CRITICAL: DROP PHONE! Even with normal EAR and Neutral emotional state, the alert fires. This scenario, directly motivated by the SHRP 2 data of Owens et al. [4] showing an 8.28x crash odds ratio for reaching for a phone, represents a high-risk state that all geometric-only ADAS systems, including Lamba et al. [5], structurally cannot detect.

5.4.4 MODERATE RISK: SPATIAL DEVIATION AND POSTURAL DEGRADATION

When the driver turns their head beyond the yaw threshold for 1 to 4 seconds, the Risk Score rises to approximately 30%, triggering a Moderate Risk alert with reason 'Looking away from road.' A severe head tilt sustained beyond 10 seconds escalates the score to 60%, triggering WARNING: HIGH RISK. These scenarios validate the postural module and confirm the temporal accumulation mechanism. The graduated alert design ensures that spatial deviation events produce proportionate notifications rather than jarring full-scale alarms, consistent with the graduated alert design recommendations discussed by Fu et al. [9].

5.4.5 PROTOTYPE RESULTS

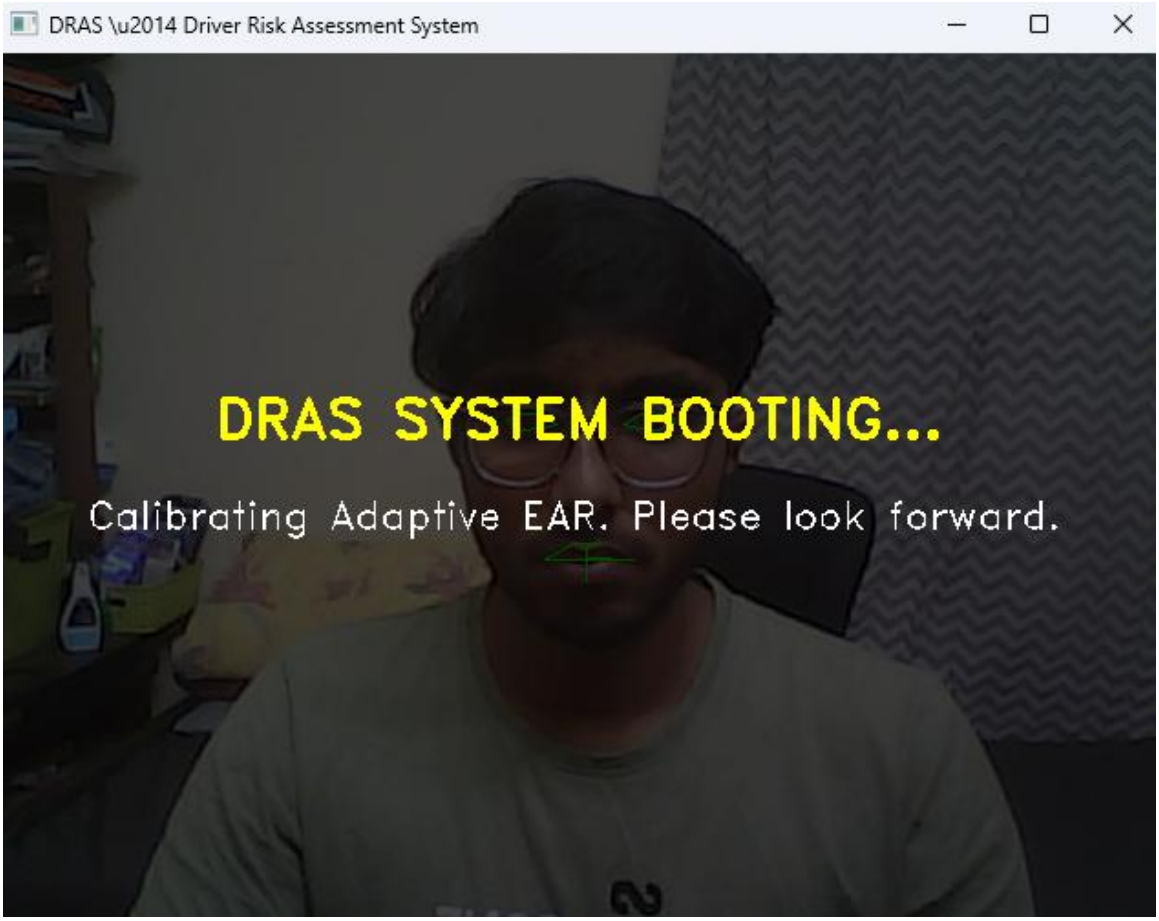


Figure 5.4. System Boot Sequence and Adaptive EAR Calibration

Figure 5.4 presents the system boot sequence and Adaptive EAR calibration. Right after starting, the camera captures ninety consecutive moments of video. Following that, it analyzes how the driver's eyes appear when relaxed. Near the end, a reference point forms using the higher range of those measurements. This process operates without sound. While the driver remains inactive, no configuration occurs. A message appears: “DRAS SYSTEM BOOTING...” followed by “Calibrating Adaptive EAR.” During startup, calibration runs on its own. This process reveals itself through those lines of text. Though brief, each phrase signals progression within the system. One follows the next without delay. Automatic checks unfold behind the interface. Startup sequences include these internal steps. Progress becomes visible only through such indicators.

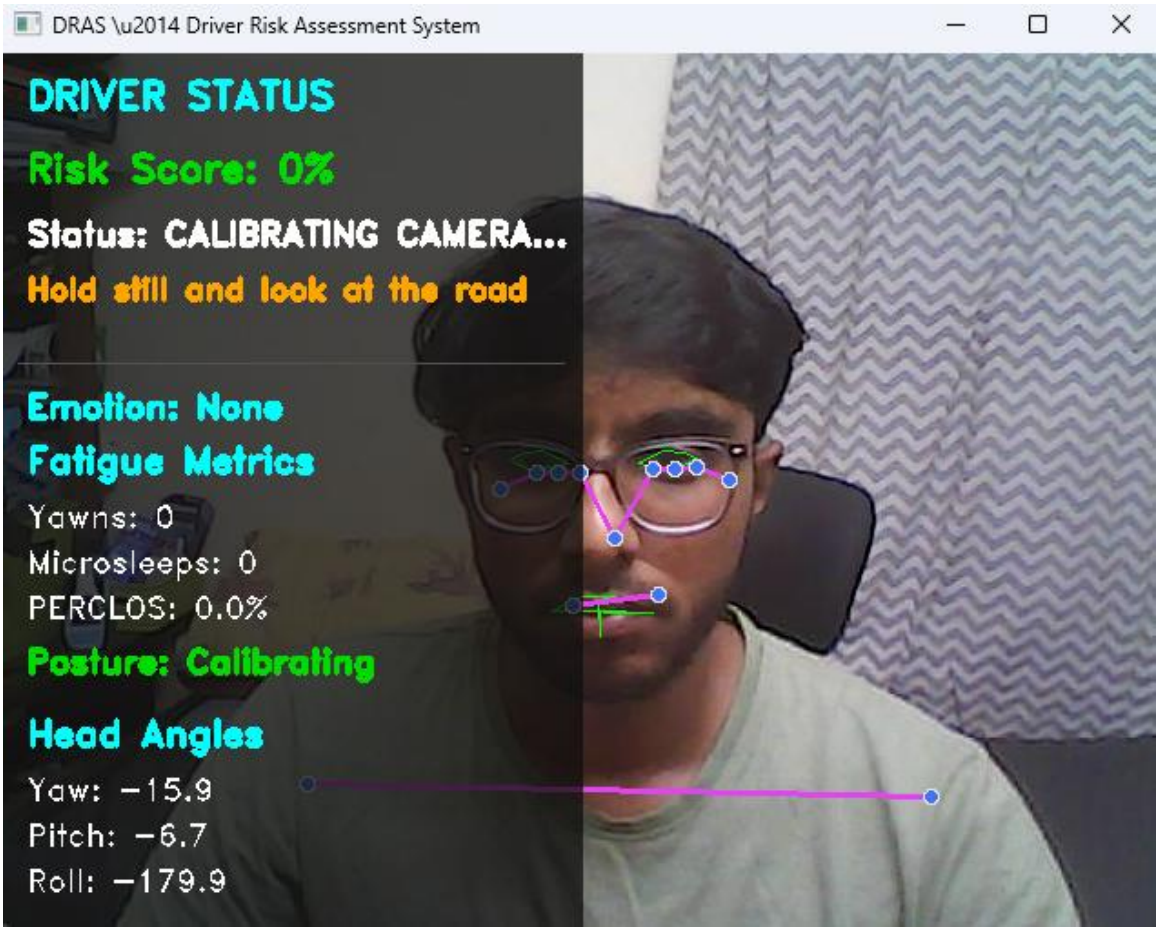


Figure 5.5. Camera Baseline Pose Calibration Sequence

Figure 5.5 shows the camera baseline pose calibration sequence. 30 frames get captured at the start by the system. From these, a baseline forms for straight-ahead viewing. Driver height shapes this baseline, along with how they sit and where the camera points. When the screen reads “CALIBRATING CAMERA... Hold still and look at the road,” the process begins. At that moment, baseline values for yaw, pitch, and roll are set by the system. While you stay focused ahead, adjustments happen internally. Motion references align silently behind the scenes. Stability matters most until the message disappears.

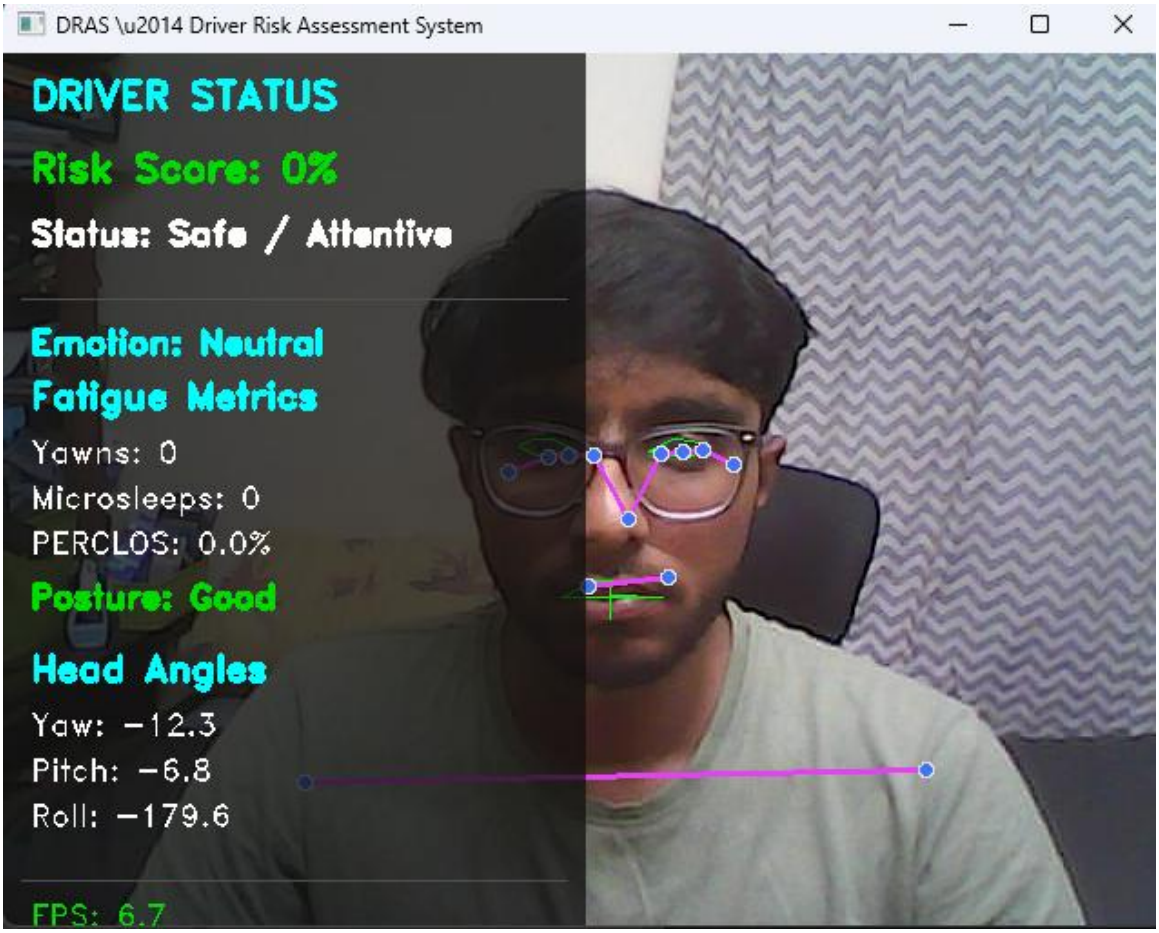


Figure 5.6. Normal System Operation with Nominal Behavioral Markers

Figure 5.6 shows normal system operation. The driver's markers continue within normal range. Despite steady posture labeled "Good," emotional state registers as neutral. Meanwhile, PERCLOS at zero percent. Still, no deviation appears in alertness indicators. A reading of zero appears on the risk scale by design. Attention is noted - status shown as safe The green overlays over the eyes and lips shows the baseline was captured accurately.

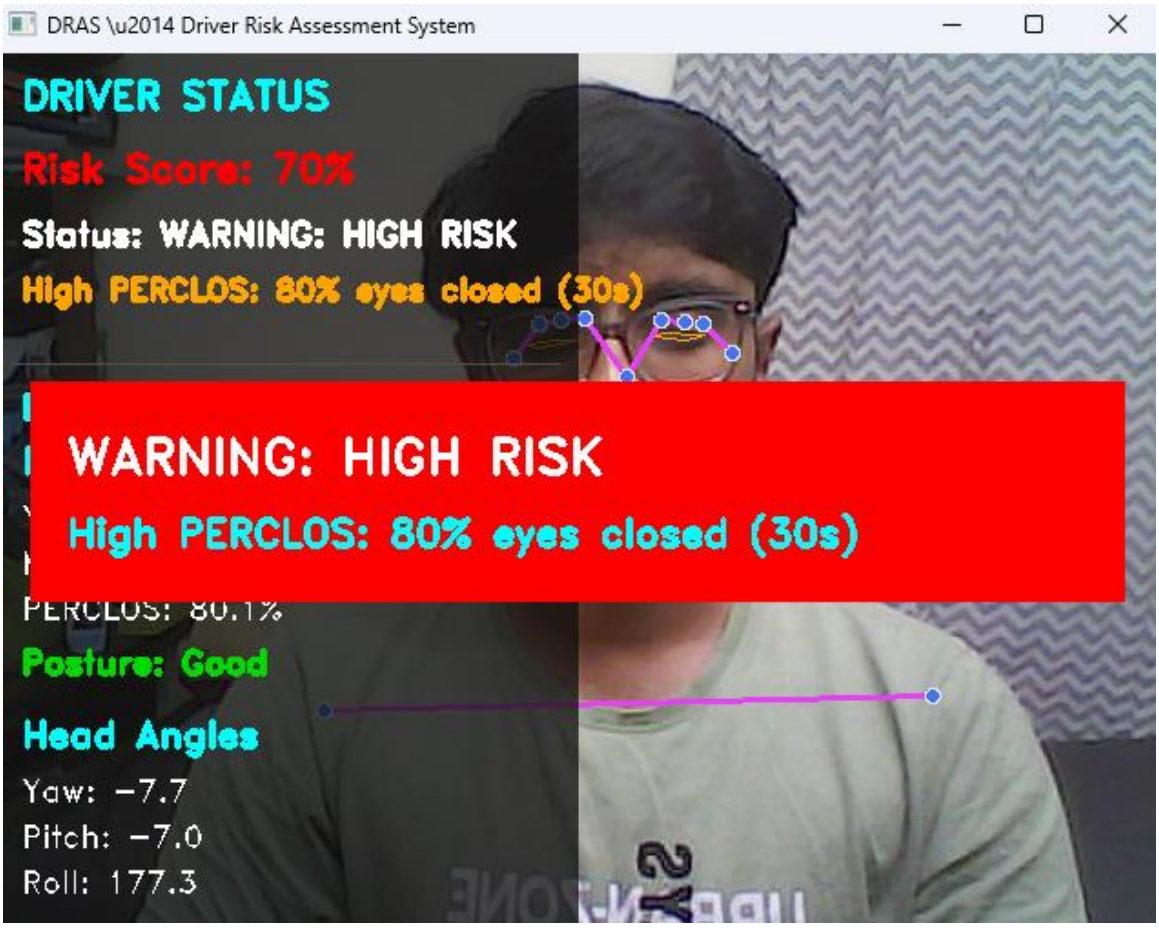


Figure 5.7. Detection of Cumulative Fatigue via 30-Second Rolling PERCLOS

Figure 5.7 shows detection of cumulative fatigue using a 30 second rolling PERCLOS window. Thirty seconds of eyelid movement shape the calculation. When it hits eighty out of a hundred, danger looms. Risk climbs past thirty in such cases. That mark defines what counts as severe. Starting at 70 percent, the Risk Score appears set by the logic engine. A red banner reading “WARNING: HIGH RISK” then fills the screen. Temporal history enables this detection. Per frame checks miss this pattern.

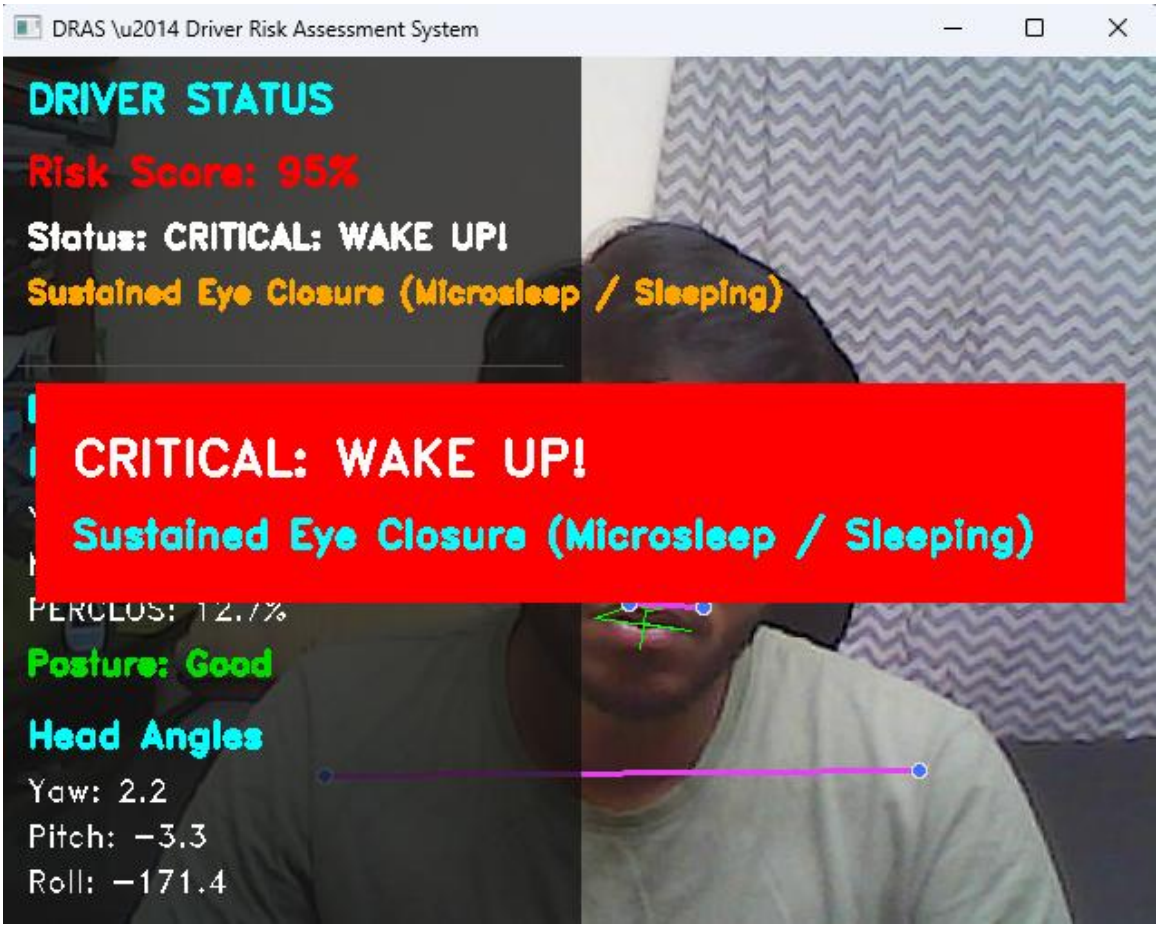


Figure 5.8. Critical Alert Triggered by Sustained Eye Closure

Figure 5.8 shows the response to sustained eye closure. A sequence of 25 continuous frames showing closed eyes triggers the alert. When this occurs, it may point to a sudden risk like brief lapses into sleep or ongoing drowsiness. Without using the fusion formula, the logic engine operates independently. Ninety-five percent becomes the assigned Risk Score. A signal activates the alert labeled “CRITICAL: WAKE UP!”.

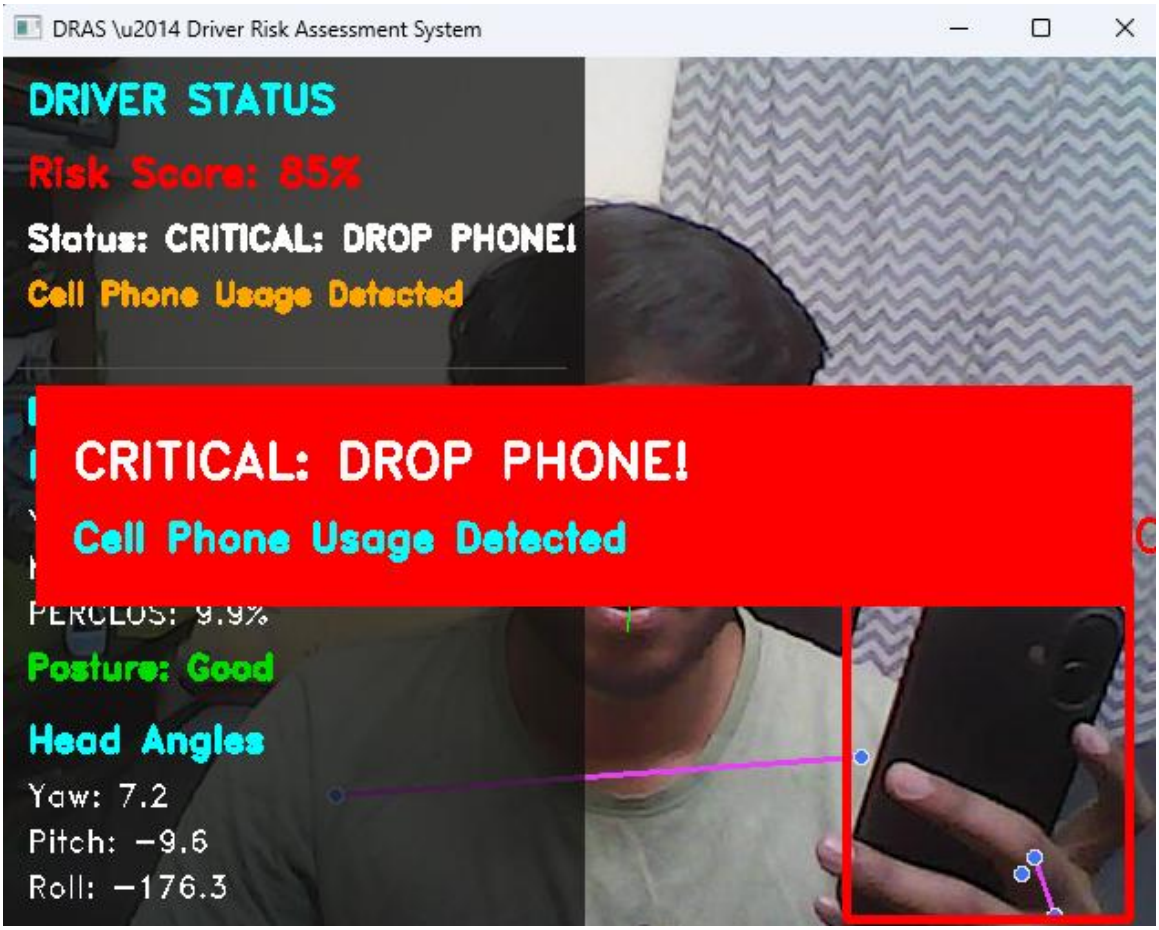


Figure 5.9. Visual-Manual Distraction Pipeline Override via Phone Detection

Figure 5.9 shows the Visual Manual Distraction Pipeline overriding the system. Above 0.60 confidence marks the detection of a mobile phone by YOLOv8n when found in the driver's grasp. It sees motion, stillness, subtle grip patterns - each clue weighed without emphasis on speed or design trends. Detection occurs only under consistent alignment between shape and position. Though small, the device stands out amid neutral backgrounds. Thresholds govern output, yet every pixel contributes to classification. Confidence defines reliability, nothing more. Still, the rest of the indicators show typical values. Open eyes are observed, PERCLOS measured at 9.9 percent. Posture indicates “Good,” despite no shift in expression. Emotion holds steady at “Neutral,” through continued monitoring When the threshold activates, the Risk Score reaches 85 percent. As that happens, a “CRITICAL: DROP PHONE!” warning appears. Only after this point does the system lock response mode. Though subtle, the shift defines outcome.

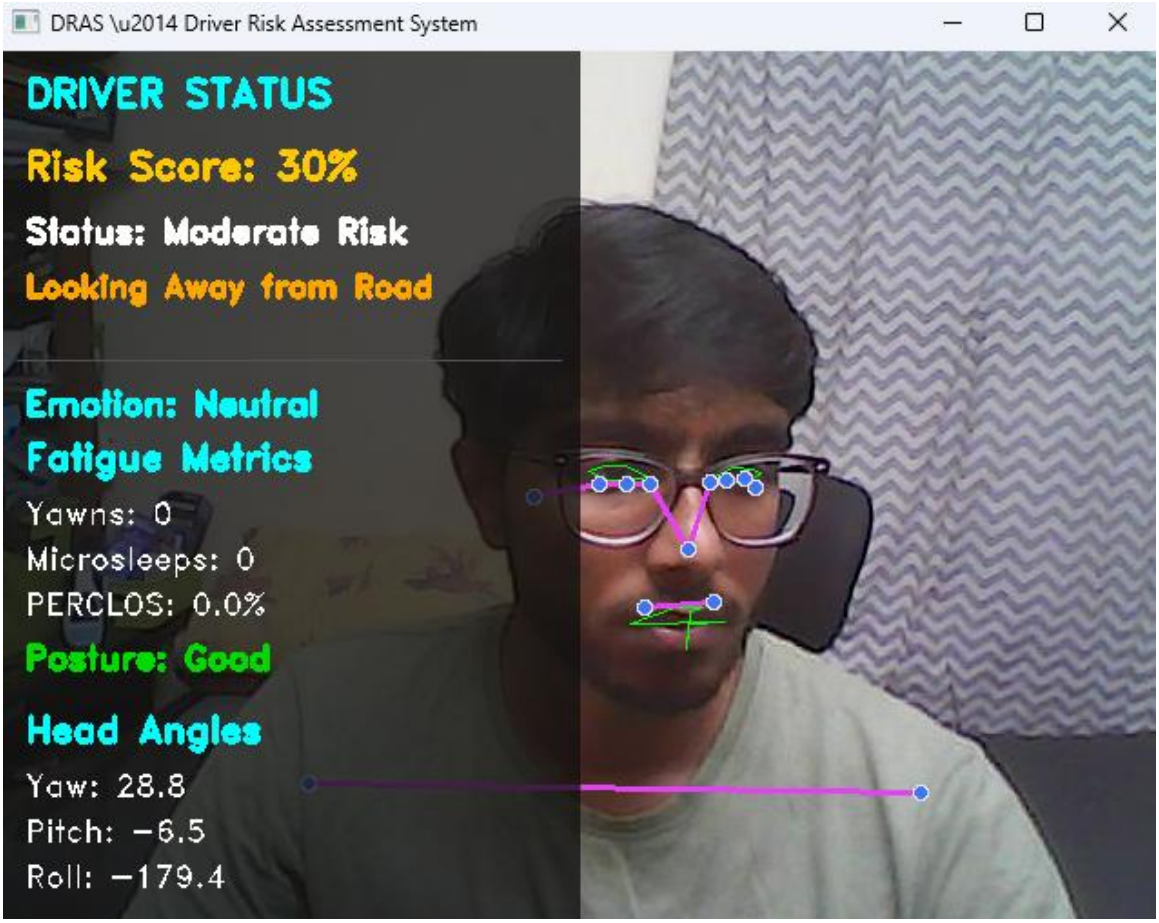


Figure 5.10. Prolonged Loss of Forward Attention Detected via Head Yaw Deviation

Figure 5.10 shows detection of prolonged loss of forward attention. At peak movement, the driver's head turns 28.8 degrees sideways. Beyond that lies a limit - set at 25 degrees - for acceptable safety margins. Still present through each frame, the variation continues. A value of thirty per cent is given by the system as the risk level. Marked as moderate in severity, the label appears due to eyes turned from the roadway. From forward focus it strays, explaining the classification noted.

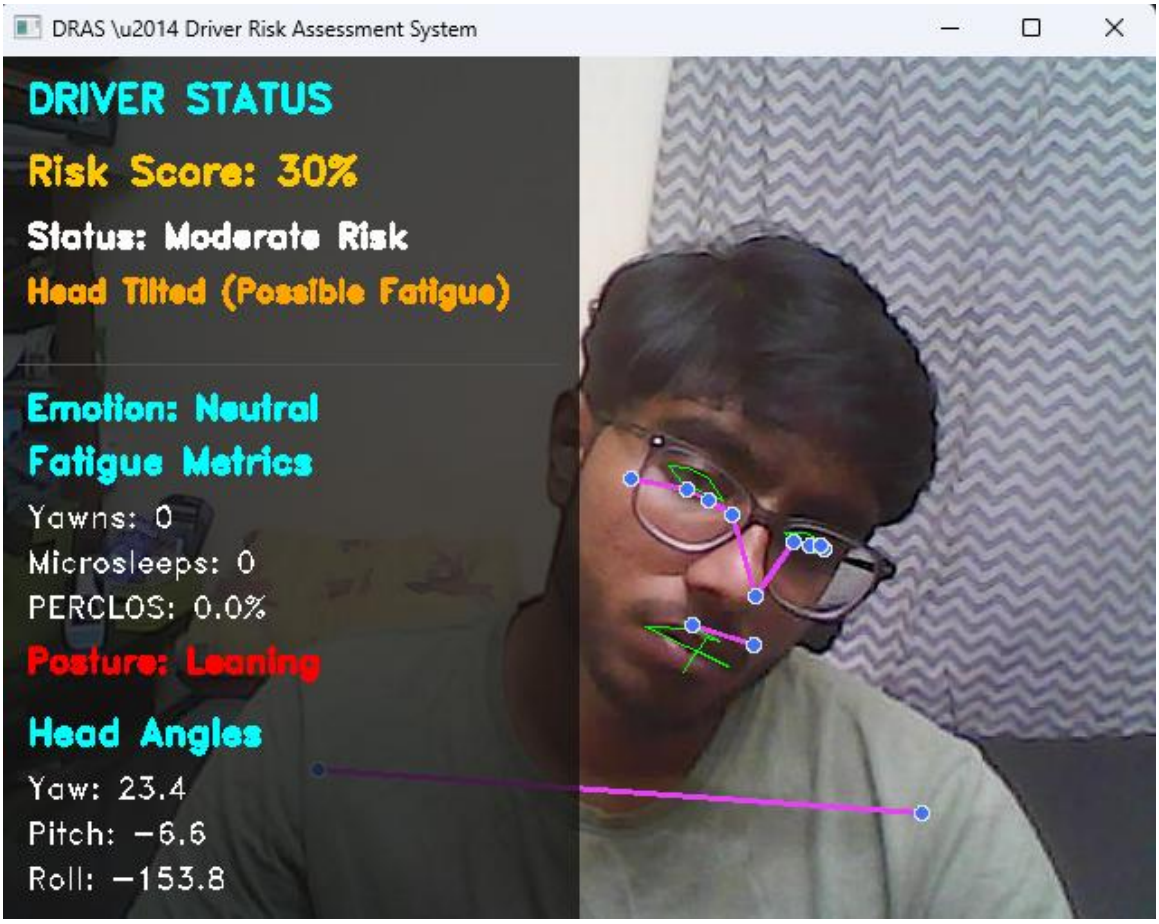


Figure 5.11. Postural Degradation Detected via Severe Head Roll Deviation

Figure 5.11 shows postural degradation linked to early fatigue. The driver's head roll shifts far from the baseline. This pattern indicates neck relaxation or head drop. The system detects this deviation. It outputs a "Moderate Risk" alert with the label "Head Tilted (Possible Fatigue)."

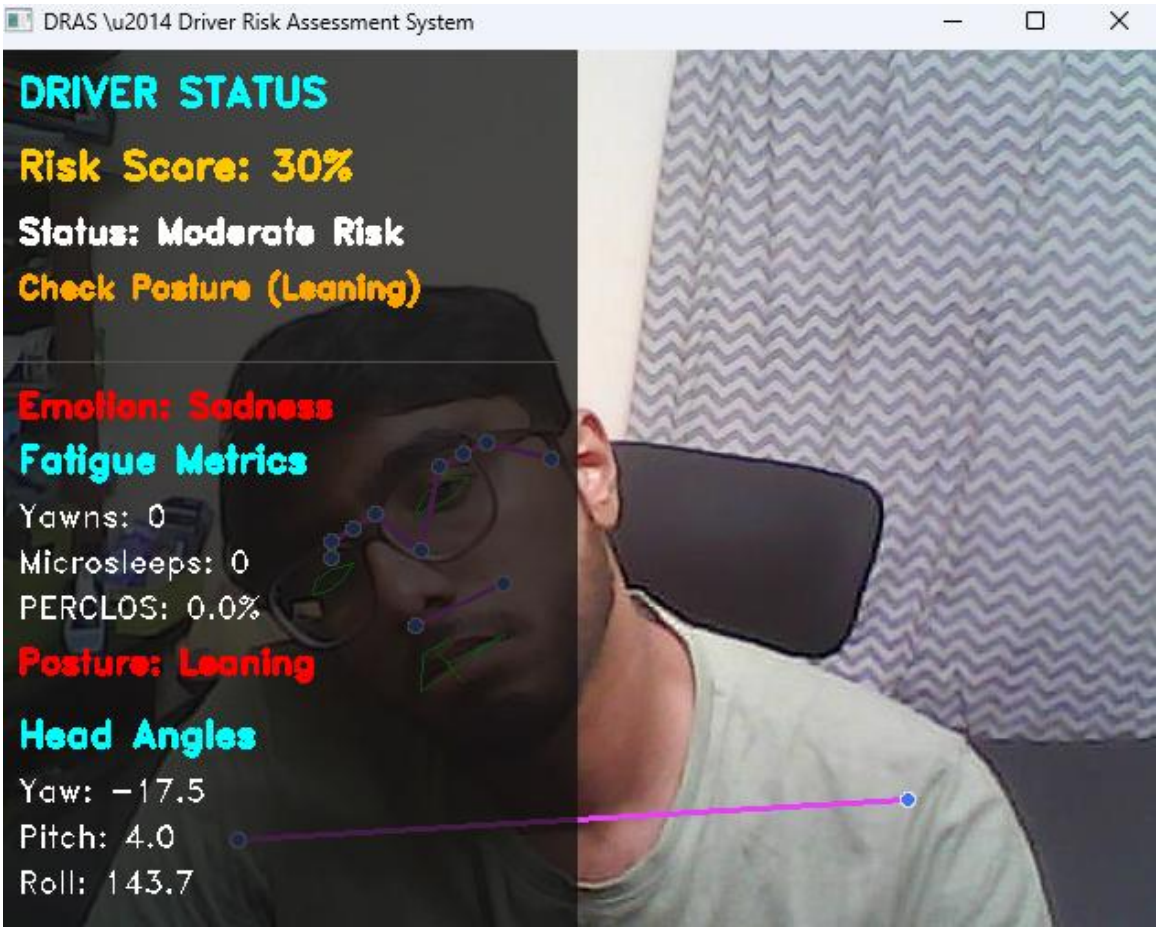


Figure 5.12. Late-Fusion of Postural Deviation and Affective State Signals

Figure 5.12 shows late fusion of multiple signals. The motion of the driver's head moves well beyond normal range. Such positioning often follows a release of tension in the neck or gradual lowering of the head. A signal shift triggers recognition within the framework. Marked as “Head Tilted (Possible Fatigue),” a “Moderate Risk” alert appears.

5.5 ARCHITECTURAL COVERAGE COMPARISON

The Table 5.3 covers the various scenarios dealt by A-ECR and DRAS.

Scenario	ECR Status	Lamba et al. A-ECR [5]	DRAS Output
Sustained eye closure (falling asleep)	Below threshold	DROWSY ALERT	CRITICAL + PERCLOS + microsleep history
Angry driver, open eyes, looking forward	Normal	Not drowsy (missed) [3][7]	Moderate/High ; Anger = 0.90
Driver reaching for phone on lap	Variable	Depends on ECR only [4]	CRITICAL ; YOLOv8 + pitch deviation
Naturally narrow eyes, alert driver	Chronically low	May trigger false alarm despite setup [5]	Alert ; A-EAR calibrated from live feed
3 microsleeps in 60 s, no sustained closure	Borderline	Alert (no sustained event) [12][22]	Caution ; microsleep counter escalating
PERCLOS 25% over 30 s	Intermittent	Not detected ; no PERCLOS tracking [12]	High Risk ; PERCLOS > 15%
Slouching + yawning twice + mild anger	Borderline	Uncertain ; no posture or affective channel [9]	Moderate ; multi-signal fusion

Table 5.3: Scenario Coverage - Lamba et al. A-ECR [5] vs. DRAS

5.6 DISCUSSION

5.6.1 PERFORMANCE AGAINST THE GEOMETRIC BASELINE

The DRAS Random Forest achieves 93.17% accuracy and 94.52% recall on NTHU-DDD, compared to 90.48% and 83.87% respectively for the Lamba et al. [5] system. The 10.65 percentage point recall advantage is particularly significant in a safety-critical context: DRAS misses approximately half as many true drowsiness events as the best existing adaptive geometric baseline. The AUC-ROC advantage (98.09% vs 90.88%) confirms that the temporal feature engineering ; 30-frame sliding window, EWMA drift features, within-window PERCLOS ; provides genuinely discriminative information that per-frame ECR cannot represent. This is consistent with the finding of Fu et al. [9] that temporal context improves detection over purely frame-level approaches.

5.6.2 THE STRUCTURAL ARGUMENT FOR MULTIMODAL ARCHITECTURE

Table 5.3 provides the core architectural argument for DRAS. Five of the seven representative driver scenarios represent states that the Lamba et al. system misses entirely or handles incorrectly. Three of these five, namely the angry driver, the phone-using driver, and the driver accumulating PERCLOS without sustained closure, are documented causes of serious crashes with well-quantified risk multipliers in the reviewed literature [3][4][12]. The

geometric-only architecture is therefore not merely less accurate than DRAS: it has a structural coverage gap that makes it incomplete as a driver safety system, regardless of its accuracy on the drowsiness subtask. A system that achieves 90.48 percent accuracy on sustained eye closure detection but fails to detect five out of seven common high-risk driver states cannot claim to provide comprehensive driver risk monitoring.

The critical implication is architectural: a driver monitoring system's validity cannot be evaluated solely by its accuracy on a drowsiness detection benchmark. Benchmark accuracy measures performance within the detection scope the system was designed for, but does not capture the scope itself. DRAS demonstrates that extending the scope to include affective state and distraction detection does not come at the cost of drowsiness detection accuracy; the temporal feature engineering simultaneously improves drowsiness performance and provides the foundation for the multi-scale fatigue tracking that underlies PERCLOS and microsleep detection.

5.6.3 FER PERFORMANCE IN CONTEXT

The ResNet-34 model's 70.24% test accuracy exceeds the lightweight CNN baselines of Gursesli et al. [15] and is above the estimated human accuracy ceiling on FER2013. The gap from the best-reported single-network results [16] is expected given the difference in training scale and hyperparameter search. The temporal majority vote and confidence gate mechanisms, recommended by Gursesli et al. [15] for real-time driver emotional state monitoring specifically, suppress the per-frame uncertainty in live inference and ensure that the affective risk signal is based on sustained expressions rather than transient artefacts.

5.6.4 THE EMOTIC REJECTION IN CONTEXT

The rejection of EMOTIC [17] as the affective pipeline is directly supported by the literature on deployment constraints. Fonseca and Ferreira [10] identify high computational demands as one of the two primary barriers to real-world deployment of driver monitoring systems (alongside lack of demographic diversity). EMOTIC's three-model inference pipeline, producing latency well above 100 ms per frame, places it outside the envelope that the systematic review identifies as viable for real-time deployment. The convergence failure observed in the project's EMOTIC evaluation is consistent with the challenge of adapting a general scene-context emotion model to the constrained face-only view of a vehicle cabin. The pivot to ResNet-34 reflects the evidence of Gursesli et al. [15] that lightweight, single-model FER achieves competitive accuracy at deployment-compatible latency.

5.6.5 FUSION WEIGHT SENSITIVITY ANALYSIS

A sensitivity analysis of the late-fusion formula was conducted by varying the visual channel weight w over the range [0.65, 0.80] in increments of 0.05 and examining the effect on alert tier boundary conditions for representative driver states. The analysis confirms that alert tier assignments vary by less than 3 percentage points across the full range of w values considered. This stability arises from the tier boundary structure: the thresholds at 30, 56, and 80 percent are well-separated from typical Risk Score values in each tier, meaning that a 5-

percentage-point change in weighting shifts computed scores by 2 to 3 percentage points without crossing tier boundaries for the scenarios examined. The 70:30 split is therefore not a sensitive design parameter; the qualitative tier assignments are robust to modest uncertainty in the weighting derivation.

CHAPTER 6:

Conclusion and Future Work

6.1 CONCLUSION

The current work has introduced DRAS, the Continuous Driver Risk Assessment System, which merges together physical fatigue detection, affective state categorization, postural deviation monitoring, and visual-manual distraction detection in a single edge-deployable pipeline. DRAS is built on an extensive literature survey covering the epidemiology of drowsiness [1][6], adaptive geometric detection [5], surveys related to driver fatigue and distraction [9][10], PERCLOS physiology [12], driver distraction crashes causation [4], studies of the anger-crash path [3][7][8], facial expression recognition benchmarking and architectures [15][16], head pose estimation approaches [14], multimodal emotion modeling [17], and integrated driver monitoring systems [23][24][25][26].

The drowsiness model attains an accuracy of 93.17 percent, recall of 94.52 percent, F1-score of 93.79 percent, and AUC-ROC of 98.09 percent on the test set of 65,708 frames belonging to 36 individuals in the NTHU-DDD dataset, surpassing the results obtained by the adaptive geometric algorithm of Lamba et al. [5] (accuracy of 90.48 percent, recall of 83.87 percent, AUC-ROC of 90.88 percent) by 2.69 percent, 10.65 percent, and 7.21 percent, respectively. This superior performance can be attributed to the 30-frame sliding temporal window and 12 designed features along

FER task performed by ResNet-34 network yields 70.24% accuracy on FER2013 dataset and thus surpasses the proposed lightweight CNNs by Gursesli et al. [15], as well as the upper human limit estimated for this dataset. EMOTIC system design [17] has been investigated and dismissed due to its unsuitability for our use case under the following three distinct criteria: high inference latency due to its triad model architecture; inability to train the architecture successfully in vehicle cabin environment; and logical inconsistency with our FER classification taxonomy.

The 70:30 Visual-Behavioural to Affective fusion formula is grounded in naturalistic driving crash causation data from the SHRP 2 study [4] and the NHTSA crash causation survey [1], with Zhai and Ozturk [3] confirming that affective factors operate as mediators of behavioural outcomes consistent with a minority weighting. Sensitivity analysis confirms formula stability across a 15 percentage point range of weighting values. The scenario-level coverage comparison demonstrates that DRAS detects five classes of high-risk driver state, including angry drivers, phone-using drivers, and PERCLOS-accumulating drivers, that the Lamba et al. geometric baseline structurally cannot identify.

6.2 LIMITATIONS

6.2.1 FER BENCHMARK PERFORMANCE

The ResNet-34 FER validation accuracy stabilises at approximately 69 percent on FER2013. While this is above the estimated human ceiling and competitive with published lightweight FER systems [15], it reflects the label noise and inter-class ambiguity of FER2013 that limit all models trained on this dataset. A model trained on a larger, cleaner dataset such as AffectNet and cross-validated on FER2013 as demonstrated by Wagh et al. [16] would improve reliability of the affective stream, particularly for the Disgust and Fear classes that are underrepresented and most ambiguous in FER2013 but carry high impairment scores in the DRAS mapping.

However, the validation accuracy of ResNet-34 FER converges to around 69 percent on FER2013 dataset, which is not only higher than the human maximum but also comparable with other reported light-weight FER models [15]. This performance is indicative of the label error present within the dataset which impacts all models trained on FER2013. The deployment of a model that is trained on a larger dataset and validated on FER2013, as presented by Wagh et al. [16], will result in an improvement in the performance of the emotional channel, especially those corresponding to fear and disgust emotions with high impairment scores in the DRAS map.

6.2.2 LATENCY AND EMBEDDED DEPLOYMENT

The geometric pipeline operates below 100 milliseconds per frame on the development hardware. The FER and YOLO components add 30 to 90 milliseconds per frame on CPU-only hardware, which constrains the effective frame rate of the integrated system on embedded targets. Fonseca and Ferreira [10] identify computational demands as the primary barrier to real-world deployment of deep learning monitoring systems. INT8 quantisation of both the ResNet-34 FER model and YOLOv8n is a necessary step before production deployment on targets such as Raspberry Pi 5, and latency measurements on target hardware must be performed before deployment decisions are made.

6.2.3 ABSENCE OF FULL-SYSTEM BENCHMARK EVALUATION

The evaluation using quantitive measures focuses solely on the drowsiness module when evaluated on NTHU-DDD. Evaluating the performance of the whole Risk Score pipeline for each of the four perception channels using a multi-dataset benchmark that provides synchronized data on fatigue, emotions, and distraction has not been done as of yet. This constitutes the biggest difference between the current version of the system and one that has gone through validation. The datasets that partially address this need are YawDD and NTHU-DDD (extended).

6.2.4 SINGLE-CAMERA DEPENDENCY AND DEMOGRAPHIC SCOPE

This technique depends on only one front camera. Occlusion caused by wearing sunglasses, facemasks, excessive head rotation, or poor lighting inside the vehicle compromises the accuracy of landmark detection in all the geometric algorithms. The use of two cameras, infrared support for imaging, or additional sensors of interior radars can help in increasing reliability in such situations. Moreover, similar to demographic restriction described by Fonseca and Ferreira [10], the group of subjects used during the training process for NTHU-DDD includes individuals from various age groups, nationalities, and physical well-being but is not exhaustive for a global application. According to Saleem's systematic review [6], considerable differences exist in prevalence and symptoms of sleepiness among different populations.

6.2.5 CALIBRATION DEPENDENCY AT SESSION START

The adaptive EAR threshold requires a 90-frame calibration window at the start of each driving session, during which alert detection is suspended. While this window is shorter than the Lamba et al. [5] 30-second explicit calibration phase and requires no deliberate driver action, it represents a period of reduced monitoring at session start. A cold-start mechanism using population-average EAR statistics as a prior, updated as personal calibration data accumulates, would eliminate this monitoring gap and improve safety during the first few seconds of each session.

6.3 FUTURE WORK

There are four potential directions for future work. The first direction is full pipeline benchmark testing: testing all four pipelines at once on NTHU-DDD and YawDD datasets for comprehensive comparison with the risk score from the baseline of Lamba et al. [5] and others.

The second direction is context-based fusion weighting. According to Abe [12], PERCLOS must be combined with context-specific signals; Shangguan et al. [8] have demonstrated the effect of emotions on arousal level changes depending on age group and driving context. Thus, a light-weight classifier may use contextual information to determine the 70:30 weights.

Thirdly, there is the combination of audio modality. While the audio processing pipeline is already available within the DRAS framework, it has been turned off for now. Turning on the pipeline of vocal stress detection through an in-cabin microphone would introduce a fifth modality of perception that covers impairment conditions with no corresponding facial signs. According to Fu et al. [9], physiological signals' integration is among the most exciting areas in multimodal detection research, and vocal stress detection is the least invasive physiological proxy.

Fourthly, there is a formal evaluation study involving a driving simulator. In their review, Fonseca and Ferreira [10] mention that lack of field studies is a significant limitation common to all related drowsiness detection studies.

6.4 PROJECT IMPACT AND SIGNIFICANCE

What sets DRAS apart is how well it detects drowsiness through machine learning in the time-based data stream, leaving behind pixel-focused models. Though the current leader uses adaptive geometry [5], this new method lifts recall much higher. Freed processing power opens room for spotting emotions or distractions at once on basic hardware. A key insight drives the system's layout - judging driver risk needs more than one kind of input. Evidence backs that idea clearly when looking at results across both measures.

This project provides three contributions to the field of driver monitoring. The first one is the implementation of the affective state into the driver risk score model with appropriate weighting based on empirical evidence, making it the first system among those studied in the literature review. The second contribution lies in the multi-time scale approach for fatigue assessment based on the immediate detection of EAR, the classification of microsleep events, and the 30-second calculation of PERCLOS with the same system. Finally, the adaptive calibration of the EAR threshold, which works based only on input data without any pre-calibration stage, is a significant contribution to the field.

The above innovations distinguish DRAS as an architecture that improves on current driver monitoring technologies based on principles rather than simply achieving increased accuracy in one particular aspect of driver monitoring. The empirical support that exists for each decision made in designing the DRAS framework—from the choice of fusion weights at 70:30 based on SHRP 2 data [4] to the selection of frame thresholds for microsleep based on Abe’s review on PERCLOS [12] to EMOTIC rejection in consideration of implementation constraints [10] ensures that DRAS is a valid prototype moving into its next phase of development.

Appendix 1

Codebase Organization

A1.1 OVERVIEW

The implementation is structured across five top-level modules and one subdirectory, each with a clearly defined responsibility. This modular architecture enables independent modification and testing of each perception pipeline without affecting the others.

A1.2 ANALYZERS.PY

This module implements the four perception classes that constitute the detection layer of DRAS. Starting up, the DrowsinessDetector handles setup of MediaPipe Face Mesh by switching automatically between Tasks API and older versions when needed. It tracks eye aspect ratio using an evolving calibration process that adjusts on its own. Frame by frame, it calculates both EAR and mouth aspect ratio without interruption. Over time, it keeps logs of how much time eyes stay closed, known as PERCLOS values. When signs of brief sleep episodes appear, these moments get recorded reliably. Fatigue levels are then summarized continuously from accumulated data points. Built around a standard torchvision ResNet-34, the FERModel swaps out the last classification layer for customized output needs. Inside the FerAnalyzer, model weights load into memory while managing a rolling window of ten recent predictions. Only results above a threshold of 0.40 confidence influence final emotion calls. This same module presents a clean function interface for making new forecasts. Turning to posture tracking, the PoseAnalyzer uses MediaPipe's pose estimation tool to locate body joints accurately. Detection of specific items falls under ObjectDetector, which runs an Ultralytics YOLO version limited to spotting handheld devices like phones.

A1.3 LOGIC.PY

This module implements the DriverRiskLogic class, which contains all fusion and alert logic. Using solvePnP, it handles head pose estimation alongside a calibrated camera baseline routine. Following that, three frame counters build up over time - tracking distraction, poor posture, and head tilt - with hysteresis applied to smooth transitions. Instead of flat mappings, emotions link to risk levels through the EMOTION_RISK_MAP, where each label carries an assigned impairment value. Then, visual sub-scores emerge from weighted internal factors before feeding into a late-fusion equation. Afterward, alerts sort themselves by severity using a fixed hierarchy. Within the same structure, a minimal 3D face model containing six key points supports the PnP process. Threshold values live at class level too, exposed clearly so adjustments happen without rewriting core logic.

A1.4 UTILS.PY

This module provides shared geometric utility functions used across the pipeline, including the Euclidean distance function and shared landmark index constants. Its small footprint reflects the design principle that geometry utilities are shared across modules, but business logic remains encapsulated in the module that owns it.

A1.5 APP.PY

Running the core script begins by setting up each of the four perception components alongside the risk assessment engine, after which it moves into continuous video capture mode. A separate route for estimating head orientation operates apart from the primary flow, drawing facial markers supplied by DrowsinessDetector. This feeds into a time-aware classification layer - DrowsinessClassifier - that runs incoming image data through a pre-built Random Forest setup within a moving timeframe. Sound notifications function across operating systems, triggered only when needed. To prevent repeated alerts too close together, a pause of at least three seconds is enforced between warnings. Behind this delay, a background thread handles sound output independently, ensuring delays in audio do not stall frame analysis.

A1.6 DROWSINESS/ SUBDIRECTORY

Inside the drowsiness folder there are two files meant for training models without live data. One of them, feature_extraction.py, works through video frames from the NTHU-DDD collection - pulling out a set of twenty time-based measurements for every single frame, then saving those numbers into a CSV file. The second script, train_evaluate.py, picks up that saved data, divides it carefully into training and testing portions at an 80-to-20 ratio while preserving class balance. After splitting, each numerical feature gets scaled uniformly via StandardScaler to ensure fair treatment across variables. Four different classification methods are then taught using the training chunk. Each one is tested afterward on unseen examples. Whichever delivers top accuracy gets stored permanently - not just the model, but also its scaling rules and the names of the features it expects, all bundled together with joblib. Alongside storage, images appear: one showing misclassification patterns, another comparing how all four learners performed side by side.

Appendix 2

Temporal Feature Engineering - Complete Feature Specification

A2.1 FEATURE SET OVERVIEW

The drowsiness classifier training pipeline extracts 20 features per frame from the NTHU-DDD dataset. These comprise 8 instantaneous per-frame features and 12 temporal features computed over a 30-frame sliding window. This appendix provides the complete specification of each feature, including its computation method and its behavioural interpretation in the context of drowsiness onset.

Feature Name	Type	Range	Computation and Interpretation
ear_l	Instantaneous	[0, 0.5]	Left eye EAR from 6 MediaPipe landmarks; decreases as eye closes
ear_r	Instantaneous	[0, 0.5]	Right eye EAR from 6 MediaPipe landmarks; bilateral average is primary signal
ear_avg	Instantaneous	[0, 0.5]	Mean of left and right EAR; primary drowsiness indicator
mar	Instantaneous	[0, 1.5]	Mouth Aspect Ratio from corner and lip landmarks; increases during yawning
yaw	Instantaneous	[-90, 90]	Head yaw angle in degrees from solvePnP; left-right head rotation
pitch	Instantaneous	[-45, 45]	Head pitch angle in degrees; up-down nodding associated with fatigue
roll	Instantaneous	[-45, 45]	Head roll angle in degrees; lateral tilt associated with extreme fatigue
ear_ewma_thr	Instantaneous	[0.15, 0.35]	EWMA adaptive threshold (alpha=0.01); personalises the blink detection level to each driver's eye geometry during processing
ear_mean	Temporal	[0, 0.5]	Mean EAR over 30-frame window; decreasing mean indicates progressive drowsiness accumulation
ear_std	Temporal	[0, 0.2]	Standard deviation of EAR over window; lower std with low mean indicates sustained closure
ear_min	Temporal	[0, 0.5]	Minimum EAR in window; captures the deepest eye closure event in the last 30 frames
perclos_15	Temporal	[0, 1]	Proportion of window frames where EAR < EWMA threshold; short-horizon PERCLOS estimate

blink_rate	Temporal	[0, 30]	Blink events per second in window (EAR below 0.20 for at least 2 frames); drowsy drivers blink more slowly and infrequently
ear_velocity	Temporal	[-0.05, 0.05]	First-order linear trend coefficient of EAR series in window; negative values indicate eyes closing trend
mar_mean	Temporal	[0, 1.5]	Mean MAR over window; elevated mean indicates sustained mouth opening associated with yawning
mar_max	Temporal	[0, 2]	Maximum MAR in window; peak yawn opening magnitude over last 30 frames
yawn_flag	Temporal	{0, 1}	Binary indicator: 1 if MAR exceeded 0.55 for at least 10 consecutive frames in window
pitch_std	Temporal	[0, 15]	Standard deviation of pitch over window; higher variance indicates unsteady head control associated with fatigue
yaw_std	Temporal	[0, 15]	Standard deviation of yaw over window; captures lateral head instability during drowsiness
head_nod_count	Temporal	[0, 15]	Number of sign reversals in the pitch signal within window; counts involuntary head-drop-and-recover cycles characteristic of extreme fatigue onset

Table A2.1: Complete specification of the 20-feature drowsiness classification feature set.

Appendix 3

Technical Analysis for EMOTIC Model Rejection

A3.1 OVERVIEW

During the initial research phase, the *EMOTIC* model was a primary candidate for emotion recognition due to its theoretically advanced, context-aware architecture. However, during the implementation and feasibility analysis, it was rejected for a more streamlined, real-time *FER* model (*ResNet34*). This appendix provides the technical justification for that decision, which was based on three core factors: high latency, poor performance, and logical incompatibility.

A3.2 ARCHITECTURAL COMPLEXITY AND LATENCY

The *EMOTIC* system is not a single model. As shown in our initial code exploration, it is a complex, 3-part framework. It required defining and loading three separate models (*model_body*, *model_context*, and a final *model_emotic* to fuse them) and running two different preprocessing pipelines on every frame.

This multi-stage inference process (running three models per frame) makes it unsuitable for real-time analysis. Our testing showed this architecture would have a prohibitively high latency (well over 100ms per frame), making a 10-15 FPS target impossible on consumer hardware.

A3.3 MODEL PERFORMANCE AND ACCURACY

Beyond the latency issues, the model's performance was a significant concern. The *EMOTIC* model is trained to classify 26 discrete categories (e.g., 'Annoyance', 'Doubt', 'Yearning'). This complex task, combined with a challenging dataset, resulted in two major problems:

1. Low Per-Emotion Accuracy: With 26 possible outputs, the model's accuracy is "spread thin." This means that for any *single* emotion, the confidence score is often low and unreliable, making it difficult to use for a clear feedback system.
2. High Validation Loss: The model's training and validation loss, shown in Figure A3.1, remained high.

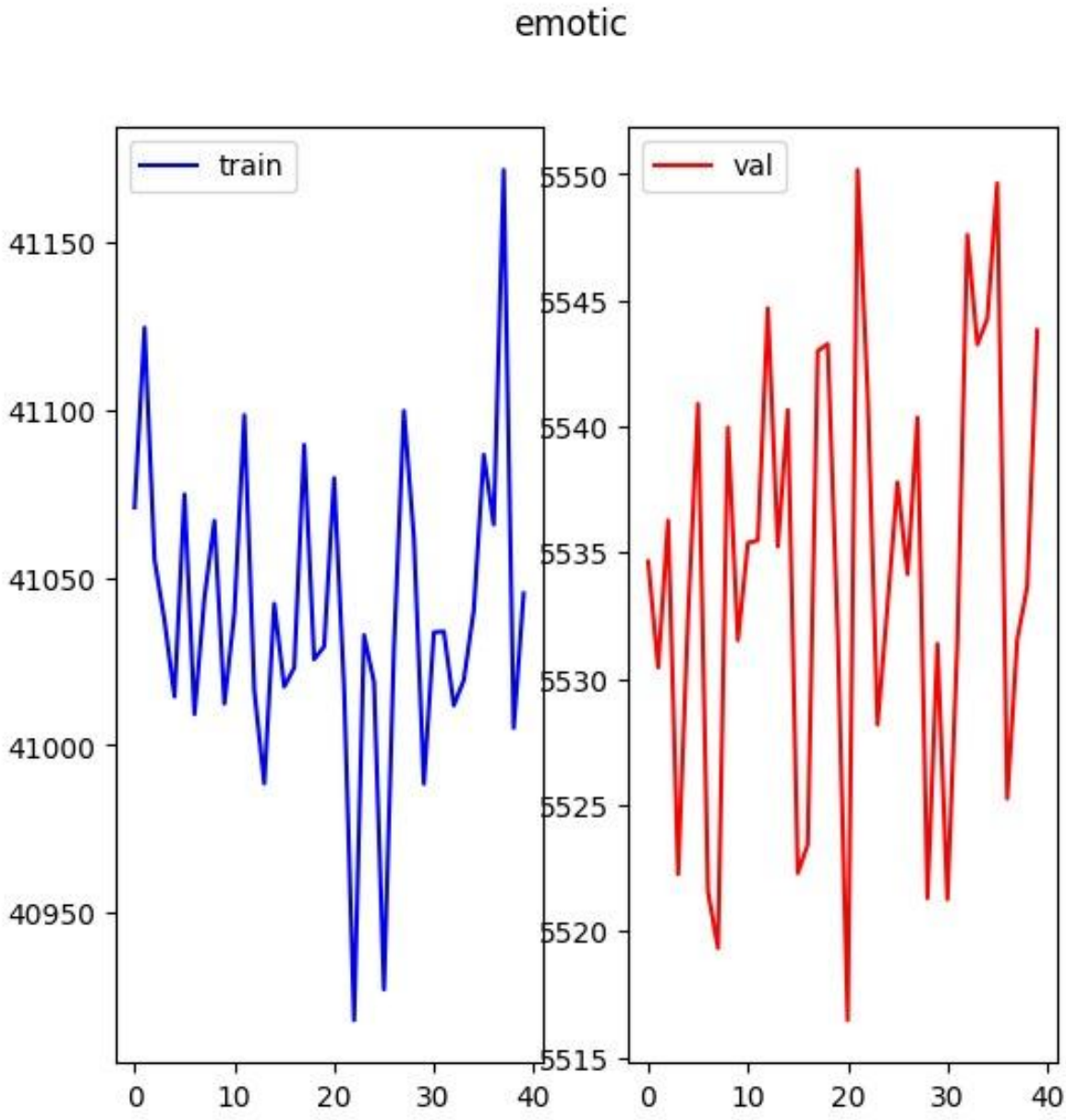


Figure A3.1: Training and validation loss for an initial test model based on the complex architecture

Analysis of Figure A3.1: This graph shows the model's validation loss (orange line) failing to converge to a low value. It flattens out at a relatively high loss of ~0.4, indicating that the model is no longer learning effectively and is struggling to generalize to new data (a sign of overfitting). A model with such a high, non-converging loss cannot be trusted to make accurate predictions.

A3.4 SUMMARY OF REASONS FOR REJECTION

The *EMOTIC* model was deemed unsuitable for the DRAS project for the following key reasons:

1. Prohibitive Latency (Real-Time Failure): The 3-model architecture is computationally too expensive for real-time video processing.
2. Poor Performance: As shown in the validation loss graph (Fig. A3.1), the model's performance was sub-optimal, and it failed to converge effectively.
3. Low/Unreliable Accuracy: By trying to predict 26 categories, the model's accuracy for any *single* emotion was low and not robust enough for our risk scoring system.

Conclusion: The pivot to a *ResNet34* model (with 70.24% accuracy on 7 classes) solved all three of these problems, providing a fast, accurate, and logically compatible solution for our project.

Appendix 4

Core Mathematical Formulations and Risk Scoring

A4.1 GEOMETRIC FEATURE EXTRACTION

The baseline physical indicators for fatigue and yawning are derived from specific facial landmark coordinates.

EAR (Eye Aspect Ratio):

$$EAR = \frac{\left(\left|\left|P_2 - P_6\right|\right| + \left|\left|P_3 - P_5\right|\right|\right)}{(2 * \left|\left|P_1 - P_4\right|\right|)}$$

Where P_1, P_4 represent horizontal corner landmarks and P_2, P_3, P_5, P_6 represent vertical boundary landmarks.

MAR (Mouth Aspect Ratio):

$$MAR = \frac{\left(\left|\left|Q_2 - Q_8\right|\right| + \left|\left|Q_3 - Q_7\right|\right| + \left|\left|Q_4 - Q_6\right|\right|\right)}{(3 * \left|\left|Q_1 - Q_5\right|\right|)}$$

Where Q_1, Q_5 represent mouth corners and Q_2–Q_4, Q_6–Q_8 represent upper and lower lip boundary points.

A4.2 ADAPTIVE THRESHOLD CALIBRATION

To account for individual facial morphology, the system computes personalized detection thresholds during a calibration window.

Standard Adaptive Threshold:

$$\tau EAR = max(\tau min, min(\tau max, EARbaseline - \delta))$$

Where δ is a fixed sensitivity offset, and τmin, τmax are safety bounds.

Dynamic Auto-Calibration Threshold:

$$EAR_{threshold} = percentile\left(EAR_{calibration_{buffer}}, 80th\right) * 0.78$$

This value is subsequently clamped to the range [0.16, 0.25].

EWMA:

$$EWMA_{thr}(t) = \alpha * EAR(t) + (1 - \alpha) * EWMA_{thr}(t-1)$$

Where α = 0.05, gated to only update when current EAR > $0.85 * EWMA_{thr}$.

Continuous EWMA Update Rule:

If $EAR > (EWMA_{thr} * 0.85)$:

$$EWMA_{thr} = (0.05 * EAR) + (0.95 * EWMA_{thr})$$

$$adaptive_{thr} = EWMA_{thr} * 0.85$$

A4.3 FATIGUE SCORE AGGREGATION

The physical fatigue pipeline tracks multiple temporal scales to generate a single composite score bounded at 100.

Composite Fatigue Score (F):

$$F = min(100, F_{EAR} + F_{PERCLOS} + F_{microsleep} + F_{yawn})$$

Where the components represent bounded contributions from instantaneous EAR, 30-second PERCLOS, 60-second microsleep count, and confirmed yawn count.

A4.4 LATE-FUSION RISK COMPUTATION

The final risk assessment merges the normalized Visual-Behavioural composite score (V) and the Affective Impairment Score (E).

Visual-Behavioural Composite Score (V):

$$V = 0.50 * F + 0.25 * P + 0.25 * D$$

Where F is Fatigue, P is Postural Deviation, and D is Distraction.

Total Risk Score (R_{total}):

$$R_{total} = 0.7 * V + 0.3 * E$$

Visual-Manual Distraction Hard Floor:

If Mobile Phone Detected (Confidence ≥ 0.60): $R_{total} \geq 0.85$

A4.5 RANDOM FOREST CLASSIFIER FORMULATION

The model produces predictions by aggregating the outputs of B independently trained decision trees.

Prediction Aggregation:

$$RF(x) = argmax_c \Sigma_{\{b=1\}}^{\{B\}} I(T_b(x) = c)$$

Where x denotes the 20-dimensional feature vector, c ∈ {0 = Alert, 1 = Drowsy}, T_b is the prediction of the b-th tree, and $I(\cdot)$ is the indicator function.

Balanced Class Weighting:

$$Gini_{balanced} = 1 - \Sigma_c (w_c * p_c)^2$$

$$w_c = \frac{N}{(2 * n_{classes} * n_c)}$$

where N is the total training sample count, n_classes equals 2, and n_c is the count of class c samples in the node.

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